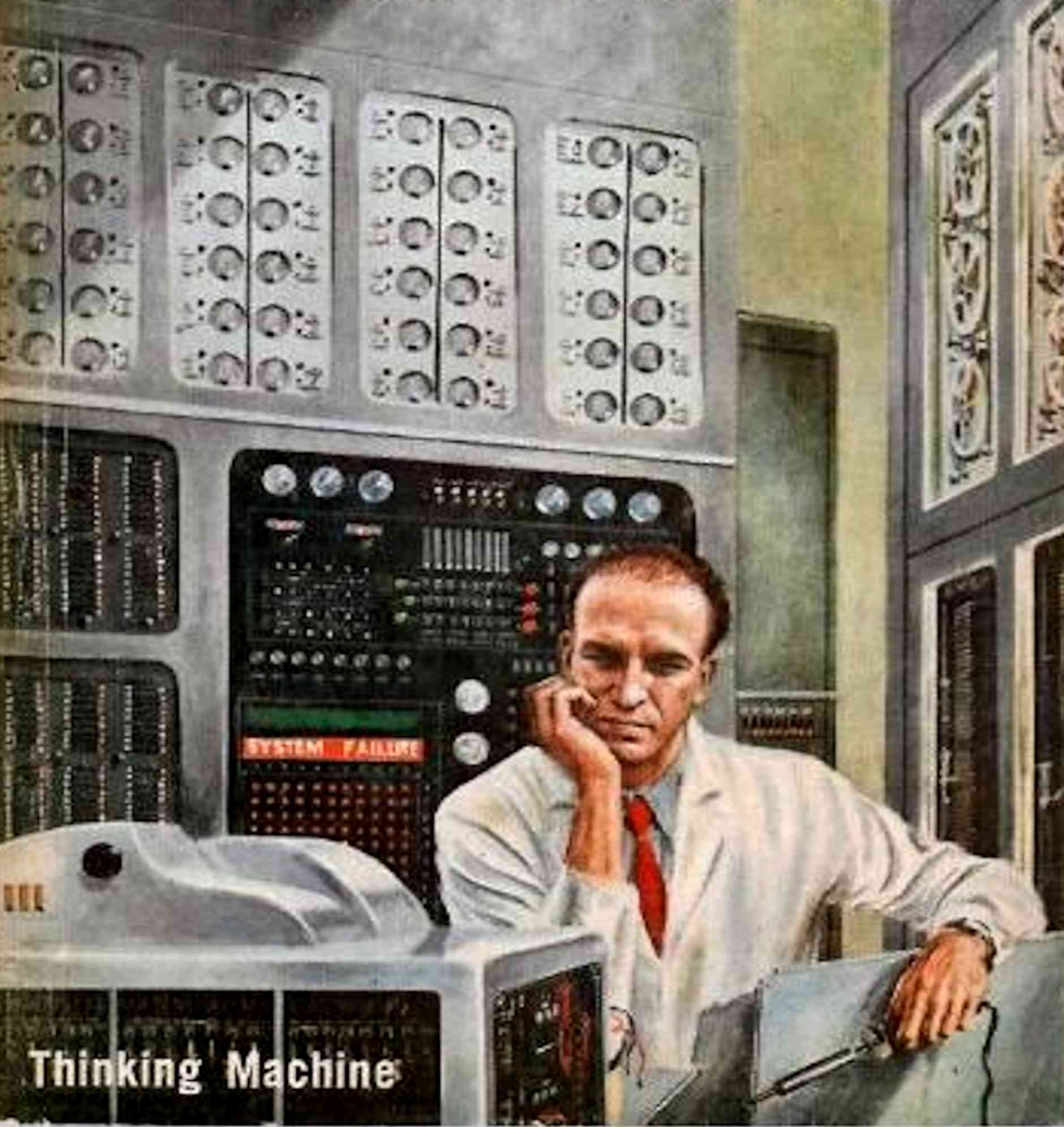


October 1957
35 Cents

Astounding SCIENCE FICTION

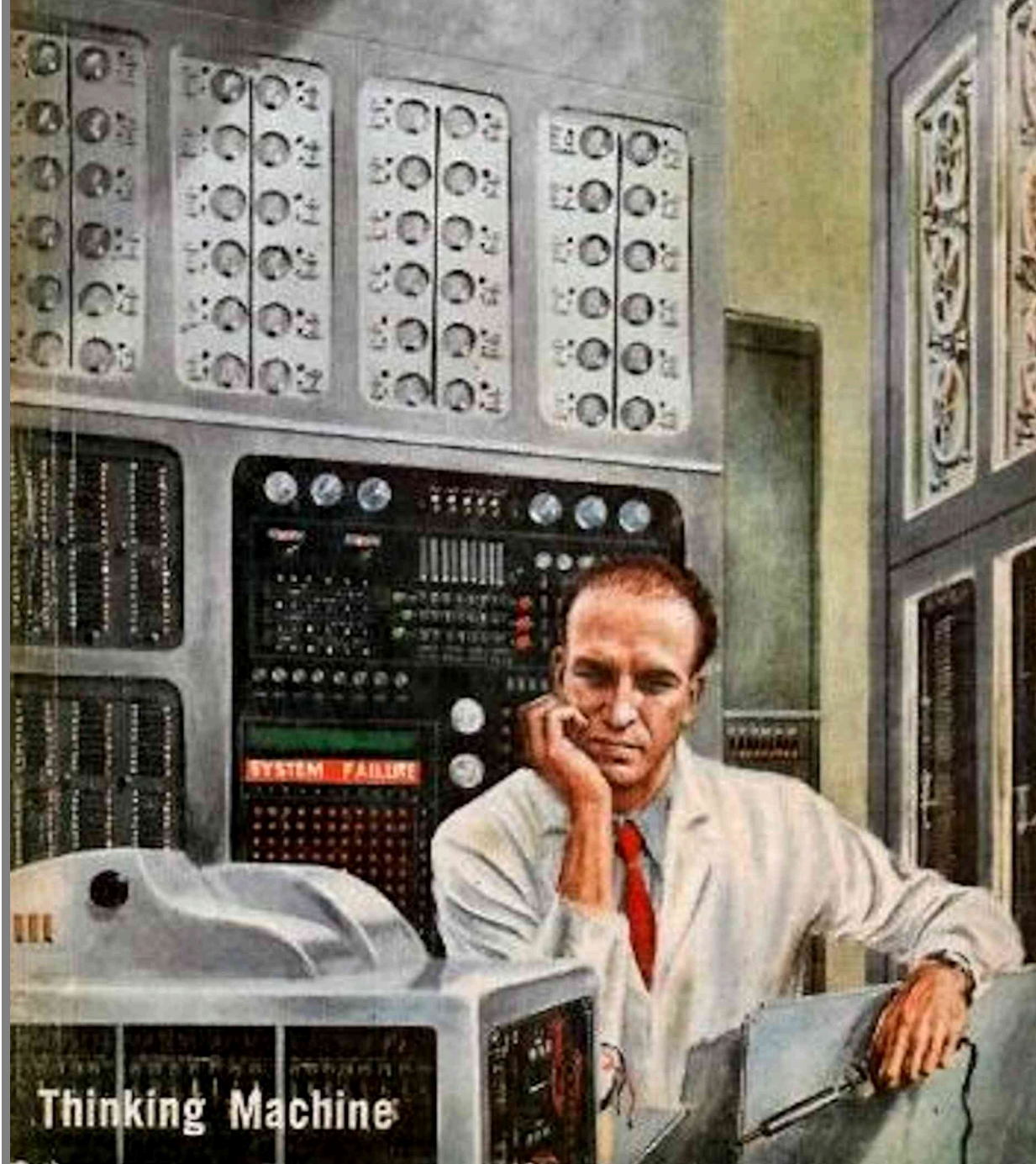


Thinking Machine

October 1957

35 Cents

Astounding SCIENCE FICTION



Thinking Machine

The Project Gutenberg eBook of The grandfathers' war

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The grandfathers' war

Author: Murray Leinster

Illustrator: H. R. Van Dongen

Release date: April 21, 2023 [eBook #70610]

Most recently updated: July 26, 2023

Language: English

Original publication: United States: Street & Smith Publications, Inc., 1957

Other information and formats: www.gutenberg.org/ebooks/70610

Credits: Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE
GRANDFATHERS' WAR ***

THE GRANDFATHERS' WAR

BY MURRAY LEINSTER

Illustrated by van Dongen

**[Transcriber's Note: This etext was produced from
Astounding Science Fiction October 1957.
Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

I.

"... No man can be fully efficient if he expects praise or appreciation for what he does. The uncertainty of this reward, as experienced, leads to modification of one's actions to increase its probability.... If a man permits himself the purpose of securing admiration, he tends to make that purpose primary and the doing of his proper work secondary. This costs human lives...."

Manual, Interstellar Medical Service. Pp. 17-18.

The little Med ship seemed absolutely motionless when the hour-off warning whirled. Then it continued to seem motionless. The background-noise tapes went on, making the small, unrelated sounds that exist unnoticed in all the places where human beings dwell, but which have to be provided in a ship in overdrive so a man don't go ship-happy from the dead stillness. The hour-off warning was notice of a change in the shape of things.

Calhoun put aside his book—the manual of the Med Service—and yawned. He got up from his bunk to tidy ship. Murgatroyd, the *tormal*, opened his eyes and regarded him drowsily, without uncoiling his furry tail from about his nose.

"I wish," said Calhoun critically, "that I could act with your realistic appraisal of facts, Murgatroyd! This is a case of no importance whatever, and you treat it as such, while I fume whenever I think of its futility. We are a token mission, Murgatroyd,—a politeness of the Med Service, which has to respond to hysterical summonses as well as sensible ones. Our time is thrown away!"

Murgatroyd blinked somnolently. Calhoun grinned wryly at him. The Med ship was a fifty-ton space-vessel—very small indeed, in these days—with a crew consisting exclusively of Calhoun and Murgatroyd the *tormal*. It was one of those little ships the Med Service tries to have call at every colonized planet at least once in four or five years. The idea is to make sure that all new developments in public health and individual medicine will spread as

widely and as fast as can be managed. There were larger Med craft to handle dangerous situations and emergencies of novel form. But all Med ships were expected to handle everything possible, if only because space travel consumed such quantities of time.

This particular journey, for example: An emergency message had come to Sector Headquarters from the planetary government of Phaedra II. Carried on a commercial vessel in overdrive at many times the speed of light, it had taken three months to reach Headquarters. And the emergency in which it asked aid was absurd. There was, said the message, a state of war between Phaedra II and Canis III. Military action against Canis III would begin very shortly. Med Service aid for injured and ill would be needed. It was therefore requested at once.

The bare idea of war, naturally, was ridiculous. There could not be war between planets. Worlds communicated with each other by spaceships, to be sure, but the Lawlor interplanetary drive would not work save in unstressed space, and of course overdrive was equally inoperable in a planet's gravitational field. So a ship setting out for the stars had to be lifted not less than five planetary diameters from the ground before it could turn on any drive of its own. Similarly, it had to be lowered an equal distance to a landing after its drive became unusable. Space travel was practical only because there were landing-grids—those huge structures of steel which used the power of a planet's ionosphere to generate the force-fields for the docking and launching of ships of space. Hence landing-grids were necessary for landings. And no world would land a hostile ship upon its surface. But a landing-grid could launch bombs or missiles as well as ships, and hence could defend its planet, absolutely. So there could be no attacks and there could be defense, so wars could not be fought.

"The whole thing's nonsense," said Calhoun. "We'll get there, and we've been three months on the way and the situation is six months old and either it's all been compromised or it's long forgotten and nobody will like being reminded of it. And we've wasted our time and talents on a thankless job that doesn't exist, and couldn't! The universe has fallen on evil days, Murgatroyd! And we are the victims!"

Murgatroyd leisurely uncurled his tail from about his nose. When Calhoun talked at such length, it meant sociability. Murgatroyd got up, and stretched,

and said, "*Chee!*" He waited. If Calhoun really meant to go in for conversation, Murgatroyd would join in. He adored pretending that he was a human. He and his kind imitated human actions as parrots imitated human speech. Murgatroyd frisked a little, to show his readiness for talk.

"*Chee-chee-chee!*" he said conversationally.

"I notice that we agree," said Calhoun. "Let's clean up."

He began those small items of housekeeping which one neglects when nothing can happen for a long time ahead. Books back in place. Files restored to order. The special-data reels Calhoun had been required to study. Calhoun made all neat and orderly against landing and possible visitors.

Presently the breakout clock indicated twenty-five minutes more in overdrive. Calhoun yawned again. As an interstellar service organization, the Med Service sometimes had to do rather foolish things. Governments run by politicians required them. Yet Med Service representatives always had to be well-informed on problems which appeared. During this journey Calhoun had been ordered to read up on the ancient insanity once called the art of war. He didn't like what he'd learned about the doings of his ancestors. He reflected that it was lucky that such things couldn't happen anymore. He yawned again.

He was strapped in the control-chair a good ten minutes before the ship was due to return to a normal state of things. He allowed himself the luxury of still another yawn. He waited.

The warning tape whirled a second time. A voice said, "*When the gong sounds, breakout will be five seconds off.*" There was a heavy, rhythmic tick-tocking. It went on and on. Then the gong and a voice said: "*Five—four—three—t—*"

It did not complete the count. There was a tearing, rending noise and the spitting of an arc. There was the smell of ozone. The Med ship bucked like a plunging horse. It came out of overdrive two seconds ahead of time. The automatic, emergency-rockets roared and it plunged this way and changed course violently and plunged that, and seemed to fight desperately against

something that frustrated every maneuver it tried. Calhoun's hair stood on end until he realized that the external-field indicator showed a terrific artificial force-field gripping the ship. He cut off the rockets as their jerkings tried to tear him out of his chair.

There was stillness. Calhoun rasped into the spacephone:

"What's going on? This is Med Ship *Esclipus Twenty*! This is a neutral vessel!" The term "neutral vessel" was new in Calhoun's vocabulary. He'd learned it while studying the manners and customs of war in overdrive. "Cut off those force-fields!"

Murgatroyd shrilled indignantly. Some erratic movement of the ship had flung him into Calhoun's bunk, where he'd held fast to a blanket with all four paws. Then another wild jerking threw him and the blanket together into a corner, where he fought to get clear, chattering bitterly the while.

"We're noncombatants!" snapped Calhoun—another new term.

A voice growled out of the spacephone speaker.

"Set up for light-beam communication," it said heavily. *"In the meantime keep silence."*

Calhoun snorted. But a Med ship was not an armed vessel. There were no armed vessels nowadays. Not in the normal course of events. But vessels of some sort had been on the watch for a ship coming to this particular place.

He thought of the word "blockade"—another part of his education in the outmoded art of war. Canis III was blockaded.

He searched for the ship that had him fast. Nothing. He stepped up the magnification of his vision-screens. Again nothing. The sun Canis flamed ahead and below, and there were suspiciously bright stars which by their coloring were probably planets. But the Med ship was still well beyond the habitable part of a sol-class sun's solar system.

Calhoun pulled a photocell out of its socket and waited. A new and very bright light winked into being. It wavered. He stuck the photocell to the screen, covering the brightness. He plugged in its cord to an audio amplifier. A dull humming sounded. Not quite as clearly as a spacephone voice, but clear enough, a voice said:

"If you are Med Ship Esclipus Twenty, answer by light-beam, quoting your orders."

Calhoun was already stabbing another button, and somewhere a signal-lamp was extruding itself from its recess in the hull. He said irritably:

"I'll show my orders, but I do not put on performances of dramatic readings! This is the devil of a business! I came here on request, to be a ministering angel or a lady with a lamp, or something equally improbable. I did not come to be snatched out of overdrive, even if you have a war on. This is a Med ship!"

The slightly blurred voice said as heavily as before:

"This is a war, yes. We expected you. We wish you to take our final warning to Canis III. Follow us to our base and you will be briefed."

Calhoun said tartly:

"Suppose you tow me! When you dragged me out of overdrive you played the devil with my power!"

Murgatroyd said, "*Chee?*" and tried to stand on his hind legs to look at the screen. Calhoun brushed him away. When acknowledgment came from the unseen other ship, and the curious cushiony drag of the towing began to be felt, he cut off the microphone to the light-beam. Then he said severely to Murgatroyd:

"What I said was not quite true, Murgatroyd. But there is a war on. To be a neutral I have to appear impressively helpless. That is what neutrality means."

But he was far from easy in his mind. Wars between worlds were flatly impossible. The facts of space travel made them unthinkable.

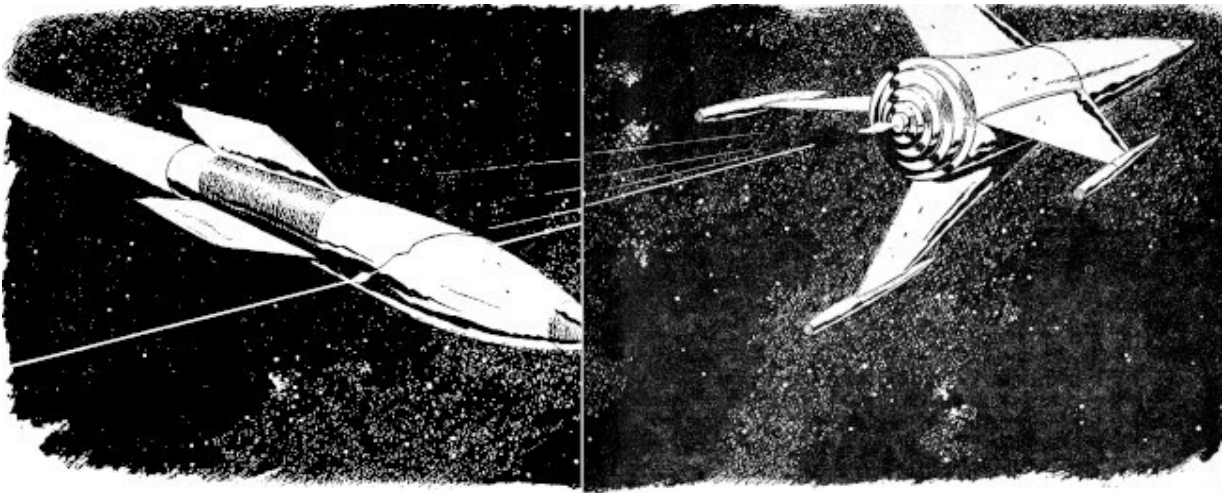
Yet there seemed to be a war. Something was happening, anyhow, which was contrary to all the facts of life in modern times. And Calhoun was involved in it. It demanded that he immediately change all his opinions and all his ideas of what he might have to do. The Med Service could not take sides in a war, of course. It had no right to help one side or the other. Its unalterable function was to prevent the needless death of human beings. So it could not help one combatant to victory. On the other hand it could not

merely stand by, tending the wounded, and by alleviating individual catastrophes allow their number to mount.

"This," said Calhoun, "is the devil!"

"*Chee!*" said Murgatroyd.

The Med ship was being towed. Calhoun had asked for it and it was being done. There should have been no way to tow him short of a physical linkage between ships. There were force-fields which could perform that function—landing-grids used them constantly—but ships did not mount them—not ordinary ships, anyhow. That fact bothered Calhoun.



"Somebody's gone to a lot of trouble," he said, scowling, "as if wars were going back into fashion and somebody was getting set to fight them. Who's got us, anyhow?"

The request for Med Service aid had come from Phaedra II. But the military action—if any—had been stated to be due on Canis III. The flaming nearby sun and its family of planets was the Canis solar system. The odds were, therefore, that he'd been snatched out of overdrive by the Phaedrian fleet. He'd been expected. They'd ordered him not to use the spacephone. The local forces wouldn't care if the planet overheard. The invaders might. Unless there were two space fleets in emptiness, jockeying for position for a battle in the void. But that was preposterous. There could be no battles in unstressed space where any ship could flick into overdrive flight in the fraction of a second!

"Murgatroyd," said Calhoun querulously, "this is all wrong! I can't make head or tail of anything! And I've got a feeling that there is something considerably more wrong than I can figure out! At a guess, it's probably a Phaedrian vessel that's hooked on to us. They didn't seem surprised when I said who I was. But—"

He checked his instrument board. He examined the screens. There were planets of the yellow sun, which now was nearly dead ahead. Calhoun saw an almost infinitely thin crescent, and knew that it was the sunward world toward which he was being towed. Actually, he didn't need a tow. He'd asked for it for no particular reason except to put whoever had stopped him in the wrong. To injure a Med ship would be improper even in war—especially in war.

His eyes went back to the external-field dial. There was a force-field gripping the ship. It was of the type used by landing-grids—a type impractical for use on shipboard. A grid to generate such a force-field had to have one foot of diameter for roughly every ten miles of range. A ship to have the range of his captor would have to be as big as a planetary landing-grid. And no planetary landing-grid could handle it.

Then Calhoun's eyes popped open and his jaw dropped.

"Murgatroyd!" he said, appalled. "Confound them, it's true! They've found a way to fight!"

Wars had not been fought for many hundreds of years, and there was no need for them now. Calhoun had only lately been studying the records of warfare in all its aspects and consequences, and as a medical man he felt outraged. Organized slaughter did not seem a sane process for arriving at political conclusions. The whole galactic culture was based upon the happy conviction that wars could never happen again. If it was possible, they probably would. Calhoun knew humanity well enough to be sure of that.

"*Chee?*" said Murgatroyd inquiringly.

"You're lucky to be a *tormal*!" Calhoun told him. "You never have to feel ashamed of your kind."

The background information he had about warfare in general made him feel skeptical in advance about the information he would presently be given. It would be what used to be called propaganda, given him under the name of

briefing. It would agree with him that wars in general were horrible, but it would most plausibly point out—with deep regret—that this particular war, fought by this particular side, was both admirable and justified.

"Which," said Calhoun darkly, "I wouldn't believe even if it were true!"

II.

"Information secured from others is invariably inaccurate in some fashion. A complete and reasoned statement of a series of events is almost necessarily trimmed and distorted and edited, or it would not appear reasonable and complete. Truly factual accounts of any series of happenings will, if honest, contain inconsistent or irrational elements. Reality is far too complex to be reduced to simple statements without much suppression of fact...."

Manual, Interstellar Medical Service. P. 25.

He was able to verify his guess about the means by which interstellar war became practical, when the Med ship was landed. Normally, a landing grid was a gigantic, squat structure of steel girders, half a mile high and a full mile in diameter. It rested upon bedrock, was cemented into unbreakable union with the substance of its planet, and tapped the ionosphere for power. When the Med ship reached the abysmal darkness of the nearest planet's shadow, there were long, long pauses in which it hung apparently motionless in space. There were occasional vast swingings, as if something reached out and made sure where it was. And Calhoun made use of his nearest-object indicator and observed that something very huge fumbled about and presently became stationary in emptiness, and then moved swiftly and assuredly down into the blackness which was the planet's night-side. When it and the planetary surface were one, the Med ship began its swift descent in the grip of landing grid-type force-fields.

It landed in the center of a grid—but not a typical grid. This was more monstrous in size than any spaceport boasted. It was not squat, either, but as tall as it was wide. As the ship descended, he saw lights in a control-system cell, midway to the ground. It was amazing but obvious. The Med ship's

captors had built a landing-grid which was itself a spaceship. It was a grid which could cross the void between stars. It could wage offensive war.

"It's infernally simple," Calhoun told Murgatroyd, distastefully. "The regular landing-grid hooks onto something in space and pulls it to the ground. This thing hooks onto something on the ground and pushes itself out into space. It'll travel by Lawlor or overdrive, and when it gets somewhere it can lock onto any part of another world and pull itself down to that and stay anchored to it. Then it can land the fleet that traveled with it. It's partly a floating dry dock and partly a landing craft, and actually it's both. It's a ready-made spaceport anywhere it chooses to land. Which means that it's the deadliest weapon in the past thousand years!"

Murgatroyd climbed on his lap and blinked wisely at the screens. They showed the surroundings of the now-grounded Med ship, standing on its tail. There were innumerable stars overhead. All about, there was the whiteness of snow. But there were lights. Ships at rest lay upon the icy ground.

"I suspect," growled Calhoun, "that I could make a dash on emergency rockets and get behind the horizon before they could catch me. But this is just a regular military base!"

He considered his recent studies of historic wars, of battles and massacres and looting and rapine. Even modern, civilized men would revert very swiftly to savagery once they had fought a battle. Enormities unthinkable at other times would occur promptly if men went back to barbarity. Such things might already be present in the minds of the crews of these spaceships.

"You and I, Murgatroyd," said Calhoun, "may be the only wholly rational men on this planet. And you aren't a man."

"*Chee!*" shrilled Murgatroyd. He seemed glad of it.

"But we have to survey the situation before we attempt anything noble and useless," Calhoun observed. "But still—what's that?"

He stared at a screen which showed lights on the ground moving toward the Med ship. They were carried by men on foot, walking on the snow. As they grew nearer it appeared that there were also weapons in the group. They were curious, ugly instruments—like sporting rifles save that their bores were impossibly large. They would be—Calhoun searched his new store of information. They would be launchers of miniature rockets, capable of firing small missiles with shaped charges which could wreck the Med ship easily.

Thirty yards off, they separated to surround the ship. A single man advanced.

"I'm going to let him in, Murgatroyd," observed Calhoun. "In war time, a man is expected to be polite to anybody with a weapon capable of blowing him up. It's one of the laws of war."

He opened both the inner and outer lock doors. The glow from inside the ship shone out on white, untrodden snow. Calhoun stood in the opening, observing that as his breath went out of the outer opening it turned to white mist.

"My name is Calhoun," he said curtly to the single dark figure still approaching. "Interstellar Medical Service. A neutral, a noncombatant, and at the moment very much annoyed by what has happened!"

A gray-bearded man with grim eyes advanced into the light from the opened port. He nodded.



"My name is Walker," he said, as curtly. "I suppose I'm the leader of this military expedition. At least, my son is the leader of the ... ah ... the enemy, which makes me the logical man to direct the attack upon them."

Calhoun did not quite believe his ears, but he pricked them up. A father and son on opposite sides would hardly have been trusted by either faction, as warfare used to be conducted. And certainly their relationship would hardly be a special qualification for leadership at any time.

He made a gesture of invitation, and the gray-bearded man climbed the ladder to the port. Somehow he did not lose the least trace of dignity in

climbing. He stepped solidly into the air lock and on into the cabin of the ship.

"If I may, I'll close the lock-doors," said Calhoun, "if your men won't misinterpret the action. It's cold outside."

The sturdy, bearded man shrugged his cape-clad shoulders.

"They'll blast your ship if you try to take off," he said. "They're in the mood to blast something!"

With the same air of massive confidence, he moved to a seat. Murgatroyd regarded him suspiciously. He ignored the little animal.

"Well?" he said impatiently.

"I'm Med Service," said Calhoun. "I can prove it. I should be neutral in whatever is happening. But I was asked for by the planetary government of Phaedra. I think it likely that your ships come from Phaedra. Your grid ship, in particular, wouldn't be needed by the local citizens. How does the war go?"

The stocky man's eyes burned.

"Are you laughing at me?" he demanded.

"I've been three months in overdrive," Calhoun reminded him. "I haven't heard anything to laugh at in longer than that. No."

"The ... our enemy," said Walker bitterly, "consider that they have won the war! But you may be able to make them realize that they have not, and they cannot. We have been foolishly patient, but we can't risk forbearance any longer. We mean to carry through to victory even if we arrive at cutting our own throats for a victory celebration! And that is not unlikely!"

Calhoun raised his eyebrows. But he nodded. His studies had told him that a war psychology was a highly emotional one.

"Our home planet Phaedra has to be evacuated," said Walker, very grimly indeed. "There are signs of instability in our sun. Five years since, we sent our older children to Canis III to build a world for all of us to move to. Our sun could burst at any time. It is certain to flare up some time—and soon! We sent our children because the place of danger was at home. We urged them to work feverishly. We sent the young women as well as the men at

the beginning, so that if our planet did crisp and melt when our sun went off there would still be children of our children to live on. When we dared—when they could feed and shelter them—we sent younger boys and girls to safety, overburdening the new colony with mouths to feed, but at the least staying ourselves where the danger was! Later we sent even the small children, as the signs of an imminent cataclysm became more threatening."

Calhoun nodded again. There were not many novas in the galaxy in any one year, even among the millions of billions of stars it held. But there had been at least one colony which had had to be shifted because of evidence of solar instability. The job in that case was not complete when the flare-up came. The evacuation of a world, though, would never be an easy task. The population had to be moved light-years of distance. Space travel takes time, even at thirty times the speed of light. Where the time of disaster—the deadline for removal—could not be known exactly, the course adopted by Phaedora was logical. Young men and women were best sent off first. They could make new homes for themselves and for others to follow them. They could work harder and longer for the purpose than any other age-group—and they would best assure the permanent survival of somebody! The new colony would have to be a place of frantic, unrelenting labor, of feverish round-the-clock endeavor, because the time-scale for working was necessarily unknown but was extremely unlikely to be enough. When they could be burdened further, younger boys and girls would be shipped—old enough to help but not to pioneer. They could be sent to safety in a partly-built colony. Later smaller children could be sent, needing care from their older contemporaries. Only at long last would the adults leave their world for the new. They would stay where the danger was until all younger ones were secure.

"But now," said Walker thickly, "our children have made their world and now they refuse to receive their parents and grandparents! They have a world of young people only, under no authority but their own. They say that we lied to them about the coming flare of Phaedora's sun: that we enslaved them and made them use their youth to build a new world we now demand

to take over! They are willing for Phaedra's sun to burst and kill the rest of us, so they can live as they please without a care for us!"

Calhoun said nothing. It is a part of medical training to recognize that information obtained from others is never wholly accurate. Conceding the facts, he would still be getting from Walker only one interpretation of them. There is an instinct in the young to become independent of adults, and an instinct in adults to be protective past all reason. There is, in one sense, always a war between the generations on all planets, not only Phaedra and Canis III. It is a conflict between instincts which themselves are necessary—and perhaps the conflict as such is necessary for some purpose of the race.

"They grew tired of the effort building the colony required," said Walker, his eyes burning as before. "So they decided to doubt its need! They sent some of their number back to Phaedra to verify our observations of the sun's behavior. Our observations! It happened that they came at a time when the disturbances in the sun were temporarily quiet. So our children decided that we were over-timid; that there was no danger to us; that we demanded too much! They refused to build more shelters and to clear and plant more land. They even refused to land more ships from Phaedra, lest we burden them with more mouths to feed! They declared for rest; for ease! They declared themselves independent of us! They disowned us! Sharper than a serpent's tooth...."

"... Is an ungrateful child," said Calhoun. "So I've heard. So you declared war."

"We did!" raged Walker. "We are men! Haven't we wives to protect? We'll fight even our children for the safety of their mothers! And we have grandchildren—on Canis III! What's happened and is happening there ... what they're doing—" He seemed to strangle on his fury. "Our children are lost to us. They've disowned us. They'd destroy us and our wives, and they destroy themselves, and they will destroy our grandchildren—We fight!"

Murgatroyd climbed into Calhoun's lap and cuddled close against him. *Tormals* are peaceful little animals The fury and the bitterness in Walker's

tone upset Murgatroyd. He took refuge from anger in closeness to Calhoun.

"So the war's between you and your children and grandchildren," observed Calhoun. "As a Med ship man—what's happened to date? How has the fighting gone? What's the state of things right now?"

"We've accomplished nothing," rasped Walker. "We've been too soft-hearted! We don't want to kill them—not even after what they've done! But they are willing to kill us! Only a week ago we sent a cruiser in to broadcast propaganda. We considered that there must be some decency left even in our children! No ship can use any drive close to a planet, of course. We sent the cruiser in on a course to form a parabolic semiorbit, riding momentum down close to atmosphere above Canopolis, where it would broadcast on standard communication frequencies and go on out to clear space again. But they used the landing-grid to strew its path with rocks and boulders. It smashed into them. Its hull was punctured in fifty places! Every man died!"

Calhoun did not change expression. This was an interview to learn the facts of a situation in which the Med Service had been asked to act. It was not an occasion in which to be horrified. He said:

"What did you expect of the Med Service when you asked for its help?"

"We thought," said Walker, very bitterly indeed, "that we would have prisoners. We prepared hospital ships to tend our children who might be hurt. We wanted every possible aid in that. No matter what our children have done—"

"Yet you have no prisoners?" asked Calhoun.

He didn't grasp this affair yet. It was too far out of the ordinary for quick judgment. Any war, in modern times, would have seemed strange enough. But a full-scale war between parents and children on a planetary scale was a little too much to grasp in all its implications in a hurry.

"We've one prisoner," said Walker scornfully. "We caught him because we hoped to do something with him. We failed. You'll take him back. We don't want him! Before you go, you will be told our plans for fighting; for the destruction, if we must, of our own children! But it is better for us to destroy them than to let them destroy our grandchildren as they are doing!"

This accusation about grandchildren did not seem conceivably true. Calhoun, however, did not question it. He said reflectively:

"You're going about this affair in a queer fashion, whether as a war or an exercise in parental discipline. Sending word of your plans to one's supposed enemy, for instance—"

Walker stood up. His cheek twitched.

"At any instant now, Phaedra's sun may go! It may have done so since we heard. And our wives—our children's mothers—are on Phaedra. If our children have murdered them by refusing them refuge, then we will have nothing left but the right—"

There was a pounding on the air-lock door.

"I'm through," rasped Walker. He went to the lock and opened the doors. "This Med man," he said to those outside, "will come and see what we've made ready. Then he'll take our prisoner back to Canis. He'll report what he knows. It may do some good."

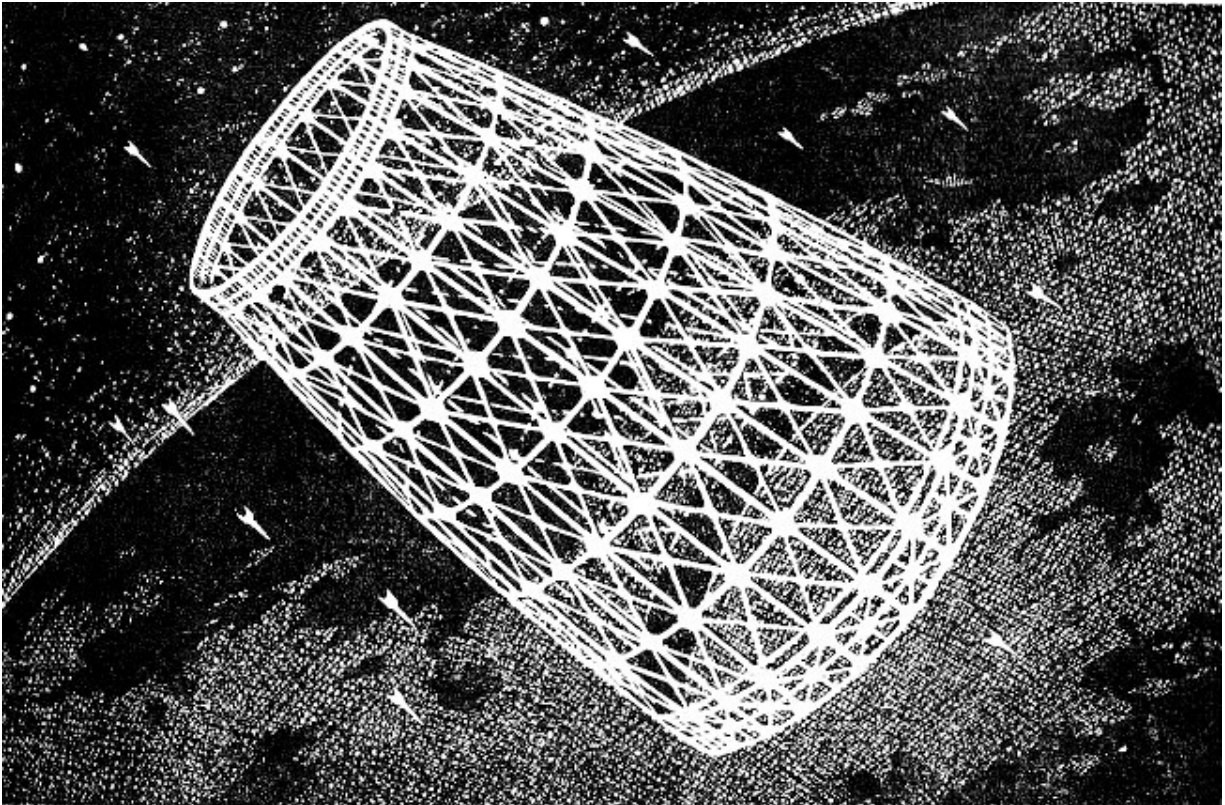
He stepped out of the air lock, flinging a command to Calhoun to follow.

Calhoun grunted to himself. He opened a cabinet and donned heavy winter garments. Murgatroyd said "*Chee!*" in alarm when it appeared that Calhoun was going to leave him. Calhoun snapped his fingers and Murgatroyd leaped up into his arms. Calhoun tucked him under his coat and followed Walker down into the snow.

This, undoubtedly, was the next planet out from the colonized Canis III. It would be Canis IV, and a very small excess of carbon dioxide in its atmosphere would keep it warmer—by the greenhouse effect—than its distance from the local sun would otherwise imply. The snow was winter snow only. This was not too cold a base for military operations against the planet next inward toward the sun.

Walker strode ahead toward the rows of spaceship hulls about the singularly spidery grip ship. It occurred to Calhoun that astrogating such a ship would be very much like handling an oversized, open-ended wastebasket. A

monstrous overdrive field would be needed, and keeping its metal above brittle-point on any really long space voyage would be difficult indeed. But it was here. It had undoubtedly lifted itself from Phaedra. It had landed itself here, and should be able to land on Canis and then let down after itself the war fleet now clustered about its base. But Calhoun tried to take comfort in the difficulty of traveling really long distances, up in the tens or twenties of light-years, with such a creation. Possibly, just possibly, warfare would still be limited to relatively nearby worlds—



"We thought," rumbled Walker, "that we might excavate shelters here, so we could bring the rest of Phaedra's population here to wait out the war—so they'd be safe if Phaedra's sun blew. But we couldn't feed them all. So we have to blast a reception for ourselves on the world our children have made!"

They came to a ship which was larger than any except the grid ship. Nearby half its hull had been opened and a gigantic tent set up against it. It was a huge machine shop. A spaceship inside was evidently the cruiser of which Walker had spoken. Calhoun could see where ragged old holes had been made in its hull. Men of middle age or older worked upon it with a

somehow dogged air. But Walker pointed to another object, almost half the size of the Med ship. Men worked on that, too. It was a missile, not man-carrying, with relatively enormous fuel-capacity for drive-rockets.

"Look that over," commanded Walker. "That's a rocket-missile, a robot fighting machine that we'll start from space with plenty of rocket fuel for maneuvering. It will fight and dodge its way down into the middle of the grid at Canopolis—which our children refuse to use to land their parents. In three days from now we use this to blast that grid and as much of Canopolis as may go with it from the blast of a megaton bomb. Then our grid ship will land and our fleet will follow it down, and we'll be aground on Canis with blast-rifles and flame and more bombs, to fight for our rightful foothold on our children's world!

"When our fighting men are landed, our ships will begin to bring in our wives from Phaedra—if they are still alive—while we fight to make them safe. We'll fight our children as if they were wild beasts—the way they've treated us! We begin this fight in just three days, when that missile is ready and tested. If they kill us—so much the better! But we'll make them do their murder with their hands, with their guns, with the weapons they've doubtless made. But they shall not murder us by disowning us! And if we have to kill them to save our grandchildren—we begin to do so in just three days! Take them that message!"

Calhoun said:

"I'm afraid they won't believe me."

"They'll learn they must!" growled Walker. Then he said abruptly: "What repairs does your ship need? We'll bring it here and repair it, and then you'll take our prisoner and carry him and your message back to his own kind—our children!"

The irony and the fury and the frustration in his tone as he said, "children," made Murgatroyd wriggle, underneath Calhoun's coat.

"I find," said Calhoun, "that all I need is power. You drained my overdrive charge when you snatched my ship out of overdrive. I've extra Duhanne cells, but one overdrive charge is a lot of power to lose."

"You'll get it back," growled Walker. "Then take the prisoner and our warning to Canis. Get them to surrender if you can."

Calhoun considered. Under his coat, Murgatroyd said "*Chee! Chee!*" in a tone of some indignation.

"Thinking of the way of my own father with me," said Calhoun wryly, "and accepting your story itself as quite true—how the devil can I make your children believe that this time you aren't bluffing? Haven't you bluffed before?"

"We've threatened," said Walker, his eyes blazing. "Yes. And we were too soft-hearted to carry out our threats. We've tried everything short of force. But the time has come when we have to be ruthless! We have our wives to consider."

"Whom," observed Calhoun, "I suspect you didn't dare have with you because they wouldn't let you actually fight, no matter what your sons and daughters did."

"But they're not here now!" raged Walker. "And nothing will stop us!"

Calhoun nodded. In view of the situation as a whole, he almost believed it of the fathers of the colonists on Canis III. But he wouldn't have believed it of his own father, regardless, and he did not think the young people of Canis would believe it of theirs. Yet there was nothing else for them to do.

It looked like he'd traveled three months in overdrive and painstakingly studied much distressing information about the ancestors of modern men, only to arrive at and witness the most heart-rending conflict in human history.

III.

"The fact that one statement agrees with another statement does not mean that both must be true. Too close an agreement may be proof that both statements are false. Conversely, conflicting statements may tend to prove each other's verity, if the conflict is in their interpretations of the facts they narrate...."

Manual, Interstellar Medical Service. P. 43.

They brought the prisoner a bare hour later. Sturdy, grizzled men had strung a line to the Med ship's power bank, and there was that small humming sound which nobody quite understands as power flowed into the Duhanne cells. The power men regarded the inside of the ship without curiosity, as if too much absorbed in private bitterness to be interested in anything else. When they had gone, a small guard brought the prisoner. Calhoun noted the expression on the faces of these men, too. They hated their prisoner. But their faces showed the deep and wrenching bitterness a man does feel when his children have abandoned him for companions he considers worthless or worse. A man hates those companions corrosively, and these men hated their prisoner. But they could not help knowing that he, also, had abandoned some other father whose feelings were like their own. So there was frustration even in their fury.

The prisoner came lightly up the ladder into the Med ship. He was a very young man, with a singularly fair complexion and a carriage at once challengingly jaunty and defiant. Calhoun estimated his age as seven years less than his own, and immediately considered him irritatingly callow and immature because of it.

"You're my jailer, eh?" said the prisoner brightly, as he entered the Med ship's cabin. "Or is this some new trick? They say they're sending me back. I doubt it!"

"It's true enough," said Calhoun. "Will you dog the air-lock door, please? Do that and we'll take off."

The young man looked at him brightly. He grinned.

"No," he said happily. "I won't."

Calhoun felt ignoble rage. There had been no great purpose in his request. There could be none in the refusal. So he took the prisoner by the collar and walked him into the air lock.

"We are going to be lifted soon," he said gently. "If the outer door isn't dogged, the air will escape from the lock. When it does, you will die. I can't save you, because if the outer door isn't dogged, all the air in the ship will go if I should try to help you. Therefore I advise you to dog the door."

He closed the inner door. He looked sick. Murgatroyd looked alarmedly at him.

"If I have to deal with that kind," Calhoun told the *tormal*, "I have to have some evidence that I mean what I say. If I don't, they'll be classing me with their fathers!"

The Med ship stirred. Calhoun glanced at the external-field dial. The mobile landing-grid was locking its force-field on. The little ship lifted. It went up and up and up. Calhoun looked sicker. The air in the lock was thinning swiftly. Two miles high. Three—

There were frantic metallic clankings. The indicator said that the outer door was dogged tight. Calhoun opened the inner door. The young man stumbled in, shockingly white and gasping for breath.

"Thanks," said Calhoun curtly.

He strapped himself in the control-chair. The vision-screens showed half the universe pure darkness and the rest a blaze of many-colored specks of light. They showed new stars appearing at the edge of the monstrous blackness. The Med ship was rising ever more swiftly. Presently the black area was not half the universe. It was a third. Then a fifth. A tenth. It was a disk of pure darkness in a glory of a myriad distant suns.

The external-field indicator dropped abruptly to zero. The Med ship was afloat in clear space. Calhoun tried the Lawlor drive, tentatively. It worked. The Med ship swung in a vast curved course out of the dark planet's shadow. There was the sun Canis, flaming in space. Calhoun made brisk observations, set a new course, and the ship sped on with an unfelt

acceleration. This was, of course, the Lawlor propulsion system, used for distances which were mere millions of miles.

When the ship was entirely on automatic control, Calhoun swung around to his unwilling companion. Murgatroyd was regarding the youthful stranger with intense curiosity. He looked at Calhoun with some apprehension.

"My name's Calhoun," Calhoun told him. "I'm Med Service. That's Murgatroyd. He's a *tormal*. Who are you and how did you get captured?"

The prisoner went instantly into a pose of jaunty defiance.

"My name is Fredericks," he said blandly. "What happens next?"

"I'm headed for Canis III," said Calhoun. "In part to land you. In part to try to do something about this war. How'd you get captured?"

"They made a raid," said young Fredericks scornfully. "They landed a rocket out in open country. We thought it was another propaganda bomb, like they've landed before—telling us we were scoundrels and such bilge. I went to see if there was anything in it good for a laugh. But it was bigger than usual. I didn't know, but men had landed in it. They jumped me. Two of them. Piled me in the rocket and it took off. Then we were picked up and brought where you landed. They tried to mind-launder me!" He laughed derisively. "Showing me science stuff proving Phaedra's sun was going to blow and cook the old home planet. Lecturing me that we were all fools on Canis, undutiful sons and so on. Saying that to kill our parents wouldn't pay."

"Would it?" asked Calhoun. "Pay, that is?"

Fredericks grinned in a superior manner.

"You're pulling more of it, huh? I don't know science, but I know they've been lying to us! Look! They sent the first gang to Canis five years ago. Didn't send equipment with them, no more than they had to. Packed the ships full of people. They were twenty years old and so on. They had to sweat! Had to sweat out ores and make equipment and try to build shelters and plant food. There were more of them arriving all the time—shipped

away from Phaedra with starvation rations so more of them could be shipped. All young people, remember! They had to sweat to keep from starving, with all the new ones coming all the time. Everybody had to pitch in the minute they got there. You never heard that, did you?"

"Yes," said Calhoun.

"They worked plenty!" said Fredericks scornfully. "Good little girls and boys! When they got nearly caught up, and figured that maybe in another month they could breathe easy, why then the old folks on Phaedra began to ship younger kids. Me among 'em! I was fifteen, and we hit Canis like a flood. There wasn't shelter, or food, or clothes to spare, but they had to feed us. So we had to help by working. And I worked! I built houses and graded streets and wrestled pipe for plumbing and sewage—the older boys were making it—and I planted ground and I chopped trees. No loafing! No fun! They piled us on Canis so fast it was root hog or die. And we rooted! Then just when we began to think that we could begin to take a breather they started dumping little kids on us! Ten-year-olds and nine-year-olds to be fed and watched. Seven-year-olds to have their noses wiped! No fun, no rest—"

He made an angry, spitting noise.

"Did they tell you that," he demanded.

"Yes," agreed Calhoun. "I heard that and more."

"All the time," raged Fredericks sullenly, "they were yelling at us that the sun back home was swelling. It was wabbling. It was throbbing like it was going to burst any minute! They kept us scared that any second the ships'd stop coming because there wasn't any more Phaedra. And we were good little boys and girls and we worked like hell. We tried to build what the kids they sent us needed, and they kept sending younger and younger kids. We got to the crack-up point. We couldn't keep it up! Night, day, every day, no fun, no loafing, nothing to do but work till you dropped, and then get up and work till you dropped again."

He stopped. Calhoun said:

"So you stopped believing it could be that urgent. You sent some messengers back to check and see. And Phaedra's sun looked perfectly normal, to them. There was no visible danger. The older people showed their scientific records, and your messengers didn't believe them. They decided they were faked. They were tired. All of you were tired. Young people need fun. You weren't having it. So when your messengers came back and said the emergency was a lie—you believed them. You believed the older people were simply dumping all their burdens on you, by lies."

"We knew it!" rasped Fredericks. "So we quit! We'd done our stuff! We were going to take time out and do some living! We were away back on having fun! We were away back on rest! We were away back just on shooting the breeze! We were behind on everything! We'd been slaves, following blueprints, digging holes and filling them up again." He stopped. "When they said all the old folks were going to move in on us, that was the finish! We're human! We've got a right to live like humans! When it came to building more houses and planting more land so more people—and old people at that—could move in to take over bossing us some more, we'd had it! We hadn't gotten anything out of the job for ourselves. If the old folks moved in, we never would! They didn't mind working us to death! To hell with them!"

"The reaction," said Calhoun, "was normal. But if one assumption was mistaken, it could still be wrong."

"What could be wrong?" demanded Fredericks angrily.

"The assumption that they lied," said Calhoun. "Maybe Phaedra's sun is getting ready to flare. Maybe your messengers were mistaken. Maybe you were told the truth."

Fredericks spat. Calhoun said:

"Will you clean that up, please?"

Fredericks gaped at him.

"Mop," said Calhoun. He gestured.

Fredericks sneered. Calhoun waited. Murgatroyd said agitatedly:

"Chee! Chee! Chee!"

Calhoun did not move. After a long time, Fredericks took the mop and pushed it negligently over the place he'd spat on.

"Thanks," said Calhoun.

He turned back to the control board. He checked his course and referred to the half-century-old Survey report on the Canis solar system. He scowled. Presently he said over his shoulder:

"How has the resting worked? Does everybody feel better?"

"Enough better," said Fredericks ominously, "so we're going to keep things the way they are! The old folks sent in a ship for a landing and we took the landing-grid and dumped rocks where it'd run into them. We're going to set up little grids all over, so we can fling bombs up—we make good bombs—if they try to land anywhere besides Canopolis. And if they do make a landing, they'll wish they hadn't! All they've dared so far is drop printed stuff calling us names and saying we've got to do what they say!"

Calhoun had the inner planet, Canis III, firmly in the center of his forward screen. He said negligently:

"How about the little kids? Most of you have quit work, you say—"

"There's not much work," bragged Fredericks. "We had to make stuff automatic as we built it, so we could all keep on making more things and not lose hands tending stuff we'd made. We got the designs from home. We do all right without working much!"

Calhoun reflected. If it were possible for any society to exist without private property, it would be this society, composed exclusively of the young. They do not want money as such. They want what it buys—now. There would be no capitalists in a world populated only by the younger generation from Phaedra. It would be an interesting sort of society, but thought for the future would be markedly lacking.

"But," said Calhoun, "what about the small children? The ones who need to be taken care of? You haven't got anything automatic to take care of them?"

"Pretty near!" Fredericks boasted. "Some of the girls like tending kids. Homely girls, mostly. But there's too many little ones. So we hooked up a psych circuit with multiple outlets for them. Some of the girls play with a couple of the kids, and that keeps the others satisfied. There was somebody studying pre-psych on Phaedra, and he was sent off with the rest to dig holes and build houses. He fixed up that trick so the girl he liked would be willing to take time off from tending kids. There's plenty of good technicians on Canis III! We can make out!"



There were evidently some very good technicians. But Calhoun began to feel sick. A psych circuit, of course, was not in itself a harmful device. It was a part of individual psychiatric equipment—not Med Service work—and its value was proved. In clinical use it permitted a psychiatrist to share the consciousness of his patient during interviews. He no longer had painfully to interpret his patient's thought-processes by what he said. He could observe the thought-processes themselves. He could trace the blocks, the mental sore spots, the ugly, not-human urges which can become obsessions.

Yes. A psych circuit was an admirable device in itself. But it was not a good thing to use for baby-tending.

There would be a great room in which hundreds of small children would sit raptly with psych-circuit receptors on their heads. They would sit quietly—very quietly—giggling to themselves, or murmuring. They would be having a very wonderful time. Nearby there would be a smaller room in which one or two other children played. There would be older girls to help these few children actually play. With what they considered adult attention every second, and with deep affection for their self-appointed nurses—why the children who actually played would have the very perfection of childhood pleasure. And their experience would be shared by—would simultaneously be known and felt by—would be the conscious and complete experience of each of the hundreds of other children tuned in on it by psych circuit. Each would feel every thrill and sensation of those who truly thrilled and experienced.

But the children so kept happy would not be kept exercised, nor stimulated to act, or think, or react for themselves. The effect of psych-circuit child-care would be that of drugs for keeping children from needing attention. The merely receiving children would lose all initiative, all purpose, all energy. They would come to wait for somebody else to play for them. And the death rate among them would be high and the health rate among those who lived would be low, and the injury to their personalities would be permanent if they played by proxy long enough.

And there was another uglier thought. In a society such as must exist on Canis III, there would be adolescents and post-adolescents who could secure incredible, fascinating pleasures for themselves—once they realized what could be done with a psych circuit.

Calhoun said evenly:

"In thirty minutes or so you can call Canopolis on spacephone. I'd like you to call ahead. Will there be anybody on duty at the grid?"

Fredericks said negligently:

"There's usually somebody hanging out there. It makes a good club. But they're always hoping the old folks will try something. If they do—there's the grid to take care of them!"

"We're landing with or without help," said Calhoun. "But if you don't call ahead and convince somebody that one of their own is returning from the wars, they might take care of us with the landing-grid."

Fredericks kept his jaunty air.

"What'll I say about you?"

"This is a Med ship," said Calhoun with precision. "According to the Interstellar Treaty Organization agreement, every planet's population can determine its government. Every planet is necessarily independent. I have nothing to do with who runs things, or who they trade or communicate with. I have nothing to do with anything but public health. But they'll have heard about Med ships. You had, hadn't you?"

"Y-yes," agreed Fredericks. "When I went to school. Before I was shipped off to here."

"Right," said Calhoun. "So you can figure out what to say."

He turned back to the control board, watching the steadily swelling gibbous disk of the planet as the Med ship drew near. Presently he reached out and cut the drive. He switched on the spacephone.

"Go ahead," he said dryly. "Talk us down or into trouble, just as you please."

IV.

"Experience directs that any assurance, at any time, that there is nothing wrong or that everything is all right, be regarded with suspicion. Certainly doctors often encounter patients who are ignorant of the nature of their trouble and its cause, and in addition have had their symptoms appear so slowly and so gradually that they were never noticed and still are not realized...."

Manual, Interstellar Medical Service. P. 68.

It was a very singular society on Canis III. After long and markedly irrelevant argument by spacephone, the Med ship went down to ground in the grip of the Canopolis landing-grid. This was managed with a deftness amounting to artistry. Whoever handled the controls did so with that impassioned perfection with which a young man can handle a mechanism he understands and worships. But it did not follow that so accomplished an operator would think beyond the perfection of performance. He came out and grinned proudly at the Med ship when it rested, light as a feather, on the clear, grassy space in the center of the city's landing-grid. He was a gangling seventeen or eighteen.

A gang—not a guard—of similar age came swaggering to interview the two in the landed spacecraft. Fredericks named where he'd been working and what he'd been doing and how he'd been taken prisoner. Nobody bothered to check his statements. But his age was almost a guarantee that he belonged on Canis. When he began his experiences as a prisoner among their enemies, all pretense of suspicion dropped away. The gang at the spaceport interjected questions, and whooped at some of his answers, and slapped each other and themselves ecstatically when he related some of the things he'd said and done in enemy hands, and talked loudly and boastfully of what they would do if the old folks tried to carry out their threats. But Calhoun observed no real preparations beyond the perfect working

condition of the grid itself. Still, that ought to defend the planet adequately—except against such a mobile spaceport as he'd been captured by, himself.

When they turned to him for added reasons to despise the older generation, Calhoun said coldly:

"If you ask me, they can take over any time they're willing to kill a few of you to clear the way. Certainly if the way you're running this particular job is a sample!"

They bristled. And Calhoun marveled at the tribal organization which had sprung up among them. What Fredericks had said in the ship began to fit neatly into place with what once had been pure anthropological theory. He'd had to learn it because a medical man must know more than diseases. He must also know the humans who have them. Oddments of culture-instinct theory popped into his memory and applied exactly to what he was discovering. The theory says that the tribal cultures from which even the most civilized social organisms stem—were not human inventions. The fundamental facts of human society exist because human instinct directs them, in exact parallel to the basic design of the social lives of ants and bees. It seemed to Calhoun that he was seeing, direct, the operation of pure instinct in the divisions of function in the society he had encountered.

Here, where a guard must be mounted against enemies, he found young warriors. They took the task because it was their instinct. It was an hereditary impulse for young men of their age to act as youthful warriors at a post of danger. There was nothing more important to them than prestige among their fellows. They did not want wisdom, or security, or families, or possessions. The instinct of their age-group directed them as specifically as successive generations of social insects are directed. They moved about in gangs. They boasted vaingloriously. They loafed conspicuously and they would take lunatic risks for no reason whatsoever.

But they would never build cities of themselves. That was the impulse of older men. In particular, the warrior age-group would be capable of immense and admirable skill in handling anything which interested them, but they would never devise automatic devices to keep a city going with next to no attention. They simply would not think so far ahead. They would fight and they would quarrel and they would brag. But if this eccentric world had survived so far, it must have additional tribal structure—it must

have some more dedicated leadership than these flamboyant young men who guarded inadequately and operated perfectly the mechanism of a spaceport facility they would never have built.

"I've got to talk to somebody higher up," said Calhoun irritably. "A chief, really—a boss. Your war with your parents isn't my affair. I'm here on Med Service business. I'm supposed to check the public health situation with the local authorities and exchange information with them. So far as I'm concerned, this is a routine job."

The statement was not altogether truthful. In a sense, preventing unnecessary deaths was routine, and in that meaning Calhoun had exactly the same purpose on Canis III as on any other planet to which he might be sent. But the health hazards here were not routine. A society is an organism. It is a whole. Instinct-theory says that it can only survive as a whole, which must be composed of such-and-such parts. This society had suffered trauma, from the predicted dissolution of Phaedra's sun. Very many lives would be lost, unnecessarily, unless the results of that traumatic experience could be healed. But Calhoun's obligation was not to be stated in such terms to these young men.

"Who is running things?" demanded Calhoun. "A man named Walker said his son was bossing things here. He was pretty bitter about it, too! Who's looking after the distribution of food, and who's assigning who to raise more, and who's seeing that the small children get fed and cared for?"

The spaceport gang looked blank. Then someone said negligently:

"We take turns getting stuff to eat, for ourselves. The ones who landed here first, mostly, go around yelling at everybody. Sometimes the things they want get done. But they're mostly married now. They live in a center over yonder."

He gestured. Calhoun accepted it as a directive.

"Can somebody take me there?" he asked.

Fredericks said grandly:

"I'll do it. Going that way, anyhow. Who's got a ground-car I can use? My girl'll be worrying about me. Been worrying because she didn't know the old folks took me prisoner."

His proposal to acquire a ground-car was greeted with derision. There were ground-cars, but those that did not need repairs were jealously reserved by individuals for themselves and their closest friends. There was squabbling. Presently a scowling young man agreed to deliver Calhoun to the general area in which the first-landed of the colonists—now grown grim and authoritative—made their homes. It was annoying to wait while so simple a matter was discussed so vociferously. By the time it was settled, Fredericks had gone off in disgust.

The scowling youth produced his ground-car. Calhoun got in. Murgatroyd, of course, was not left behind. And the car was magnificent in polish and performance. Lavish effort and real ability had gone into its grooming and adjustment. With a spinning of wheels, it shot into immediate high speed. The dark-browed younglings drove with hair-raising recklessness and expertness. He traversed the city in minutes, and at a speed which allowed Calhoun only glimpses. But he could see that it was almost unoccupied.

Canopolis had been built by the youth of Phaedra to the designs of their elders for the reception of immigrants from the mother planet. It had been put up in frantic haste and used only as a receiving-depot. It had needed impassioned and dedicated labor, and sustained and exhausting concentration to get it and the rest of the colonial facilities built against a deadline of doom. But now its builders were fed up with it. It was practically empty. The last arrivals had scattered to places where food supplies were nearer and a more satisfactory way of life was possible. There were broken windows and spattered walls. There was untidiness everywhere. But there had been great pains taken in the building. Some partly-completed enterprises showed highly competent workmanship.

Then the city ended and was a giant pile of structures which fell swiftly behind. The highways were improvised. They could be made more perfect later. Across the horizon there were jerry-built villages—temporary by design, because there had been such desperate need for so many of them so soon.

The ground-car came to a stop with a screaming of brakes at the edge of such a jerry-built group of small houses. A woman ran to hiding. A man ran into view. Another, and another, and another. They came ominously toward the car.

"Hop out," said the scowling driver. He grinned faintly. "They don't want me here. But I stirred 'em up, eh?"

Calhoun stepped out of the ground-car. It whirled on one pair of wheels and sped back to the city, its driver turning to make a derisive gesture at the men who had appeared. They were still quite young men—younger than Calhoun. They looked at him steadily.

He growled to himself. Then he called:

"I'm looking for somebody named Walker. He's supposed to be top man here."

A tense young man said sardonically:

"I'm Walker. But I'm not tops. Where'd you come from? With a Med Service uniform and a *tormal* on your shoulder you're not one of us! Have you come to argue that we ought to give in to Phaedra?"

Calhoun snorted.

"I've a message that an attack from space is due in three days, but that's all from Phaedra. I'm a Med Service man. How's the health situation? How are you equipped for doctors and such? How about hospitals? How's the death rate?"

The younger Walker grinned savagely.

"This is a new colony. I doubt there are a hundred people on the planet over twenty-five. How many doctors would there be in a population like ours? I don't think there is a death rate. Do you know how we came to be here?"

"Your father told me," said Calhoun, "at the military base on the next planet out. They're getting ready for an attack—and they asked me to warn you about it. Three days from now."

Young Walker ground his teeth.

"They won't dare attack. We'll smash them if they do. They lied to us! Worked us to death—"

"And no death rate?" asked Calhoun.

The younger man knitted his brows.

"There's no use your arguing with us. This is our world! We made it and we're keeping it! They made fools of us long enough!"

"And you've no health problems at all?"

The sardonic young man hesitated. One of the others said coldly:

"Make him happy. Let him talk to the women. They're worried about some of the kids."

Calhoun breathed a private sigh of relief. These relatively mature young men were the first-landed colonists. They'd had the hardest of all the tasks put upon the younger generation by the adults of Phaedra. They'd had the most back-breaking labor and the most urgent responsibilities. They'd been worked and stressed to the breaking-point. They'd finally arrived at a decision of desperation.

But apparently things could be worse. It is the custom, everywhere, for women to make themselves into whatever is most attractive to men. Young girls, in particular, will adopt any tradition which is approved of by their prospective husbands. And in a society to be formed brand-new, appalling new traditions could be started. But they hadn't. Deep-rooted instincts still worked. Women—young women—and girls appeared still to feel concern for young children which were not even their own. And Fredericks' story—

"By all means," agreed Calhoun. "If there's something wrong with the health of children—"

Young Walker gestured and turned back toward the houses. He scowled as he walked. Presently he said defensively:

"You probably noticed there aren't many people in the city?"

"Yes," said Calhoun. "I noticed."

"We're not fully organized yet," said Walker, more defensively still. "We weren't doing anything but build. We've got to get organized before we'll

have a regular economic system. Some of the later-comers don't know anything but building. When they're ready for it, the city will be occupied. We'll have as sound a system for production and distribution of goods as anywhere else. But we've just finished a revolution. In a sense we're still in it. But presently this world will be pretty much like any other—only better."

"I see," said Calhoun.

"Most people live in the little settlements, like this—close to the crops we grow. People raise their own food, and so on. In a way you may think we're primitive, but we've got some good technicians! When they get over not having to work for the old folks and finish making things just for themselves—we'll do all right. After all, we weren't trained to make a complete world. Just to make a world for the older people on Phaedra to take over! But we've taken it over for ourselves!"

"Yes," agreed Calhoun politely.

"We'll work out the other things," said young Walker truculently. "We'll have money, and credit, and hiring each other and so on. Right now defending ourselves is the top thing in everybody's mind."

"Yes," agreed Calhoun again. He was regarded as not quite an enemy, but he was not accepted as wholly neutral.

"The older ones of us are married," Walker said firmly, "and we feel responsibility, and we're keeping things pretty well in line. We were lied to, though, and we resent it. And we aren't letting in the old people to try to run us, when we've proved we can make and run a world ourselves!"

Calhoun said nothing. They reached a house. Walker turned to enter it, with a gesture for Calhoun to accompany him. Calhoun halted.

"Just a moment. The person who drove me here—when he turned up, at least one woman ran away and you men came out ... well ... pretty pugnaciously."

Walker flushed angrily.

"I said we had technicians. Some of them made a gadget to help take care of the children. That's harmless. But they want to use it to ... to spy on older people with it. On us! Invasion of privacy. We don't like ... well ... they try to set up psych circuits near our homes. They ... think it's fun to ... know what people say and do—"

"Psych circuits can be useful," observed Calhoun, "or they can be pretty monstrous. On the other hand—"

"No decent man would do it!" snapped young Walker. "And no girl would have anything to do with anybody—But there are some crazy fools—"

"You have described," said Calhoun dryly, "a criminal class. Only instead of stealing other people's possessions they want to steal their sensations. Peeping-Tom stuff, eavesdropping on what other people feel about those they care for, as well as what they do and say. In a way it's a delinquency problem, isn't it?"

"There can't be a civilization without problems," said Walker. "But we're going to—" He opened a door. "My wife works with the kids the old people dumped on us. This way."

He motioned Calhoun inside the house. It was one of the shelters built during the frenzied building program designed to make an emergency refuge for the population of a planet. It was the roughest of machine-tool constructions. The floors were not finished. The walls were not smooth. The equipment showed. But there had been attempts to do something about the crudity. Colors had been used to try to make it homelike.

When a girl came in from the next room, Calhoun understood completely. She was a little younger than her husband, but not much. She regarded Calhoun with that anxiety with which a housekeeper always regards an unexpected visitor, hoping he will not notice defects. This young wife had those feminine instincts which are much older than tradition. Obligations and loyalties may be thrown aside, but a housewife's idea of her role is unchangeable.

"This is a Med Service man," said Walker briefly, indicating Calhoun. "I told him there was a health problem about some of the children." To Calhoun he said curtly: "This is my wife Elsa."

Murgatroyd said "*Chee!*" from where he clung to Calhoun's neck. He was suddenly reassured. He scrambled down to the floor. Elsa smiled at him.

"He's tame!" she said delightedly. "Maybe—"

Calhoun extended his hand. She took it. Murgatroyd, swaggering, extended his own black paw. Instead of conflict and hatred, here, Murgatroyd seemed to sense an amiable sociability such as he was used to. He felt more at home. He began zestfully to act like the human being he liked to pretend he was.

"He's delightful!" said the girl. "May I show him to Jak?"

Young Walker said:

"Elsa's been helping with the smaller kids. She says there's something the matter that she doesn't understand. She has one of the kids here. Bring him, Elsa."

She vanished. A moment later she brought in a small boy. He was probably six or seven. She carried him. He was thin. His eyes were bright, but he was completely passive in her arms. She put him down in a chair and he looked about alertly enough, but he simply did not move. He saw Murgatroyd, and beamed. Murgatroyd went over to the human who was near his own size. Swaggering, he offered his paw once more. The boy giggled, but his hand lay in his lap.

"He doesn't do anything!" said Elsa distressedly. "His muscles work, but he doesn't work them! He just sits and waits for things to be done for him! He acts as if he'd lost the idea of moving, or doing anything at all! And—it's beginning to show up among the other children! They just sit! They're bright enough ... they see and understand—but they just sit!"

Calhoun examined the boy. His expression grew carefully impassive. But he winced as he touched the pipestem arms and legs. What muscles were there were almost like dough.

When he straightened up, despite himself his mouth was awry. Young Walker's wife said anxiously:

"Do you know what's the matter with him?"

"Basically," said Calhoun with a sort of desperate irony, "he's in revolt. As the rest of you are in revolt against Phaedra, he's in revolt against you. You needed rest you didn't get and recreation you couldn't have and something besides back-breaking labor under a load that grew heavier minute by minute for years. You revolted, and you've a fine justification for the war in which you're engaged. But he has needed something he hasn't had, too. So he's revolting against his lack—as you did—and he's dying as you will presently do from exactly the same final cause."

Walker frowned ominously.

"I don't understand what you're saying!" he said harshly.

Calhoun moistened his lips.

"I spoke unprofessionally. The real cause of his present troubles and your future ones is that a social system has been shattered. The pieces can't live by themselves. And I don't know what medical measures can be taken to cure an injured civilization. As a medical man, I may be whipped. But I'd better check—Did I say, by the way, that the war fleet from Phaedra is going to attack in just three days?"

V.

"... Truth is the accord of an idea with a thing. Very often an individual fails to discover the truth about some matter because he neglects to become informed about something. But even more often, the truth is never found out because somebody refuses to entertain an idea...."

Manual, Interstellar Medical Service. Pp. 101-2.

On the first day, Calhoun went grimly to the *creches* that had been set up by the first-arrived young colonists when ships began to discharge really young children at the landing-grid in Canopolis. The *creches* were not too much like orphanages, of course, but the younger generation of Phaedra had been put in a very rough situation by the adults. If the time of the imminent solar explosion had been known, the matter could have been better handled. Actually, the explosion had been delayed—to date—for nearly five years from the discovery that it must occur. If that much leeway could have been predicted, older men and many machines would have been sent at first. But the bursting could not be computed. It was a matter of probability. Such-and-such unrhythmic variables must inevitably coincide sooner or later. When they did—final and ultimate catastrophe. The sun would flare terribly and destroy all life in its solar system. It could be calculated that the odds were even that the explosion would happen within one year, two to one within two, and five to one within three. The odds were enormous against Phaedra surviving as long as it had. The people of the mother-world had had a highly improbable break.

But in cold common sense they'd done the sensible thing. They'd tried to save those of their children who could take care of themselves first, and added others as they dared. But the burden on the young colonists had been monstrous. Even adults would have tended to grow warped with such pressure to mine, build, plough, and sow, as was put upon the youngsters. There had never been more than barely enough of food—and more mouths were always on the way. There had never been extra shelter, and younger

and ever younger cargoes were constantly arriving, each needing more of shelter and of care than the ones before. And there was the world of adults still to be provided for.

Calhoun met the girls who had devoted themselves to the quasi-orphaned children. They bore themselves with rather touching airs of authority among the smaller children. But they were capable of ferocity, on occasion. They had the need, sometimes, not to defend their charges but themselves against the clumsily romantic advances of loutish teen-agers who considered themselves fascinating.

They had done very well.

The small children were exactly what Calhoun had anticipated—in every way. The small boy Calhoun had seen first was an extreme case, but the results of play by proxy were visible everywhere. Calhoun constrainedly inspected one after another of the children shelters. He was anxiously watched by the sober young faces of the nurses. But they giggled when Murgatroyd tried to go through Calhoun's actions of taking temperatures and the like. He had to be stopped when he attempted to take a throat-swabbing which Calhoun had said was pure routine.

After the fourth such inspection he said to Elsa:

"I don't need to see any more. What's happened to the boys the same age as these girl nurses—the thirteen and fourteen and fifteen-year-olds?"

Elsa said uncomfortably:

"They're mostly off in the wilds. They hunt and fish and pioneer. They don't care about girls. Some of them grow things.... I don't think there'd be enough food if they didn't, even though we're not getting anybody new to feed."

Calhoun nodded. In all the cities of the galaxy, small children of both sexes were to be seen everywhere, and girls of the early teen-ages, and adults. But the boys' age-group he'd mentioned always made itself invisible. It congregated in groups away from the public eye, and engaged in adventurous games and quite futile explorations. It was socially quite self-sufficient everywhere.

"Your husband," said Calhoun, carefully impassive, "had better try to gather in some of them. As I remember it, they're capable of a rather admirable romantic idea of duty—for a while. We're going to need some romanticists presently."

Elsa had faith in Calhoun now, because he seemed concerned about the children. She said unhappily:

"Do you really think the ... old people will attack? I've grown older since I've been here. Those of us who came first are almost like the people on Phaedra—some ways. The younger people are inclined to be suspicious of us because we ... try to guide them."

"If you're confiding that you think there may be two sides to this war," Calhoun told her, "you are quite right. But see what your husband can do about gathering some of the hunting-and-fishing members of the community. I've got to get back to my ship."

He got himself driven back to the landing-grid. Walker did not drive him, but another of the now-suspect men of twenty-five or so, from the shelter village of the first-landed colonists. He was one of those who'd worked with Walker from the beginning and with him had been most embittered. Now he found himself almost a member of an older generation. He was still bitter against the people of Phaedra, but—

"This whole business is a mess!" he said darkly as he drove through the nearly deserted city toward the landing-grid. "We've got to figure out a way to organize things that'll be better than the old way. But no organization at all is no good, either! We've got some tough young characters who like it this way, but they've got to be tamed down!"

Calhoun had his own unsettling suspicions. There have always been splendid ideas of social systems which will make earthly paradises for their inhabitants. Here, by happen-chance, there had come to be a world inhabited only by the young. He tried to put aside, for the moment, what he was unhappily sure he'd find out back at the ship. He tried to think about

this seemingly perfect opportunity for a new and better organization of human lives.

But he couldn't believe in it. Culture-instinct theory is pretty well worked out. The Med Service considered it proven that the basic pattern of human societies is instinctual rather than evolved by trial and error. The individual human being passes through a series of instinct-patterns which fit him at different times to perform different functions in a social organization which can vary but never change its kind. It has to make use of the successive functions its members are driven by instinct to perform. If it does not use its members, or give scope to their instincts, it cannot survive. The more lethal attempts at novel societies tried not only to make all their members alike, but tried to make them all alike at all ages. Which could not work.

Calhoun thought unhappily of the tests he meant to make in the Med ship. As the ground-car swerved into the great open center of the grid, he said:

"My job is doing Med Service. I can't advise you how to plan a new world. If I could, I wouldn't. But whoever does have authority here had better think about some very immediate troubles."

"We'll fight if Phaedra attacks!" said the driver darkly. "They'll never get to ground alive, and if they do—they'll wish they hadn't!"

"I wasn't thinking of Phaedra," said Calhoun.

The car stopped close by the Med ship. He got out. There had been attempts to enter the ship in his absence. The gang which occupied the control building and in theory protected Canis III against attack from the sky had tried to satisfy their curiosity about the little ship. They'd even used torches on the metal. But they hadn't gotten in.

Calhoun did. Murgatroyd chattered shrilly when he was put down. He scampered relievedly about the cabin, plainly rejoicing at being once more in familiar surroundings. Calhoun paid no attention. He closed and dogged the air-lock door. He switched on the spacephone and said shortly:

"Med Ship *Esclipus Twenty* calling Phaedrian fleet! Med Ship *Esclipus Twenty* calling—"

The loud-speaker fairly deafened him as somebody yelled into another spacephone mike in the grid-control building.

"Hey! You in the ship! Stop that! No talking with the enemy!"

Calhoun turned down the incoming volume and said patiently:

"Med Ship *Esclipus Twenty* calling fleet from Phaedra. Come in, fleet from Phaedra! Med Ship *Esclipus Twenty* calling—"

There was a chorus of yellings from the nearby building. The motley, swaggering, self-appointed landing-grid guard had tried to break into the ship out of curiosity, but they were vastly indignant when Calhoun did something of which they disapproved. They made it impossible for him to have heard a reply from the space fleet presumably overhead. But after a moment someone in the control house evidently elbowed the others aside and shouted:

"You! Keep that up and we'll smash you! We've got the grid to do it with, too!"

Calhoun said curtly:

"Med ship to Control. I've something to tell you. Suppose you listen. But not on spacephone. Have your best grid technician come outside and then I shall tell him by speaker."

He snapped off the spacephone and watched. The control building fairly erupted indignant youths. After a moment he saw the gangling one who'd grinned so proudly when the Med ship was landed with absolute perfection. The others shouted and scowled at the ship.

Calhoun threw on the outside speaker—normally used for communication with a ground crew before lifting.

"I'm set," said Calhoun coldly, "for overdrive travel. My Duhanne cells are charged to the limit. If you try to form a force-field around this ship, I'll dump half a dozen overdrive charges into it in one jolt that will blow every coil you've got! And then how'll you fight the ships from Phaedra? I'm

going to talk to them on spacephone. Listen in if you like. Monitor it. But don't try to bother me!"

He threw on the spacephone again and patiently resumed his calling:

"Med ship *Esclipus Twenty* calling fleet from Phaedra! Med ship calling fleet from Phaedra—"

He saw violent argument outside the grid's control building. Some of the young figures raged. But the youth who'd handled the grid so professionally raged at them. Calhoun hadn't made an idle threat. A grid-field could be blown out. A grid could be made useless by one of the ships it handled. When a ship like Calhoun's went into overdrive, it put out something like four ounces of pure energy to form a field in which it could travel past the speed of light. In terms of horsepower or kilowatt hours, so much force would be meaningless. It was too big. It was a quantity of energy whose mass was close to four ounces. When the ship broke out of overdrive, that power was largely returned to storage. The loss was negligible, compared to the total. But, turned loose into a grid's force-field, even three or four such charges would work havoc with the grid's equipment.

Calhoun got an answer from emptiness just as the members of the group by the control building shouted each other down and went inside to listen with bitter unease and suspicion to his talk with the enemy.

"*Phaedra fleet calling*," said a growling voice in the spacephone speaker. "What do you want?"

"To exercise my authority as a Med Service officer," said Calhoun heavily. "I warn you that I now declare this planet under quarantine. All contact with it from space is forbidden until health hazards here are under control. You will inform all other spacecraft and any other spaceport you may contact of this quarantine. Message ends."

Silence. A long silence. The growling voice rasped:

"What's that? Repeat it!"

Calhoun repeated it. He switched off the phone and unpacked the throat-swabbings he'd made at the four children's shelters in turn. He opened up his laboratory equipment. He put a dilution of one throat-swabbing into a culture slide that allowed living organisms to be examined as they multiplied. He began to check his highly specific suspicions. Presently he was testing them with minute traces of various antibodies. He made rough but reasonably certain identifications. His expression grew very, very sober. He took another swab sample and put it through the same process. A third, and fourth, and fifth, and tenth. He looked very grim.

It was sunset outside when there was a hammering on the ship's hull. He switched on a microphone and speaker.

"What do you want?" he asked flatly.

The angry voice of young Walker came from the gathering darkness. The screens showed a dozen or more inhabitants of Canis III milling angrily about him. Some were of the young-warrior age. They engaged in bitter argument. But the younger Walker, and four or five with him, faced the ship with ominous quietness.

"What's this nonsense about quarantine?" demanded Walker harshly, from outside. *"Not that we've space-commerce to lose, but what does it mean?"*

"It means," Calhoun told him, "that your brave new world rates as a slum. You've kept kids quiet with psych circuits, and they haven't eaten properly and haven't exercised at all. They're weak from malnutrition and feeble from not doing their own playing. They're like slum children used to be in past ages. Here on Canis you're about ready to wipe yourselves out. You may have done it."

"You're crazy!" snapped Walker. But he was upset.

"In the four shelters I visited," Calhoun said drearily, "I spotted four cases of early diphtheria, two of typhoid, three of scarlet fever and measles, and samples of nearly any other disease you care to name. The kids have been developing those diseases out of weakness and from the reservoir of infections we humans always carry with us. They'd reached the contagious stage before I saw them—but all the kids are kept so quiet that nobody noticed that they were sick. They've certainly spread to each other and their nurses, and therefore out into your general population, all the infections

needed for a first-rate multiple epidemic. And you've no doctors, no antibiotics—not even injectors to administer shots with if you had them."

"*You're crazy!*" cried young Walker. "*Crazy! Isn't this a Phaedra trick to make us give in?*"

"Phaedra's trick," said Calhoun more drearily than before, "is an atom bomb they're going to drop into this landing-grid—I suspect quarantine or no quarantine—in just two days more. Considering the total situation, I don't think that matters."

VI.

"... The most difficult of enterprises is to secure the co-operation of others in enterprises those others did not think of first...."

Manual, Interstellar Medical Service. P. 189.

Calhoun worked all night, tending and inspecting the culture incubators which were part of the Med ship's technical equipment. In the children's shelters, he'd swabbed throats. In the ship, he'd diluted the swabbings and examined them microscopically. He'd been depressingly assured of his very worst fears as a medical man—all of which could have been worked out in detail from the psych circuit system of child care boastfully described by Fredericks. He could have written out his present results in advance from a glance at the child Jak shown him by the younger Walker's wife. But he hated to find that objective information agreed with what he would have predicted by theory.

In every human body there are always germs. The process of good health is in part a continual combat with slight and unnoticed infections. Because of victories over small invasions, a human body acquires defenses against larger invasions of contagion. Without such constant small victories, a body ceases to keep its defenses strong against beach-heads of infection. Yet malnutrition or even exhaustion can weaken a body once admirably equipped for this sort of guerilla warfare.

If an undernourished child fails to win one skirmish, he can become overwhelmed by a contagion the same child would never have known about had he only been a little stronger. But, overwhelmed, he is a sporadic case of disease—a case not traceable to another clinical case. And then he is the origin of an epidemic. In slum conditions a disease not known in years can arise and spread like wild-fire. With the best of intentions and great technical ingenuity, the younger-generation colonists of Canis III had made

that process inevitable among the younger children who were their last-imposed burden. The children were under-exercised, under-stimulated, and hence under par in appetite and nutrition. And it is an axiom of the Med Service that a single underfed child can endanger an entire planet.

Calhoun proved the fact with appalling certainty. His cultures astounded even him. But by dawn he had applied Murgatroyd's special genetic abilities to them. Murgatroyd said "*Chee!*" in a protesting tone when Calhoun did what was necessary at that small patch on his flank which was quite insensitive. But then Murgatroyd shook himself and admiringly scowled back at Calhoun, imitating the intent and worried air that Calhoun wore. Then he followed Calhoun about in high good spirits, strutting on his hind legs, man-fashion, and pretending to set out imaginary apparatus as Calhoun did, long ahead of time for what he hoped would occur.

Presently Murgatroyd tired—a little quicker than usual—and went to sleep. Calhoun bent over him and counted his respiration and heart beat. Murgatroyd slept on. Calhoun gnawed his fingers in anxious expectation.

He'd come on this assignment with some resentment because he thought it foolish. He'd carried on with increasing dismay as he found it not absurd. Now he watched over Murgatroyd with the emotional concern a medical man feels when lives depend upon his professional efficiency, but that efficiency depends on something beyond his control. Murgatroyd was that something this time—but there was one other.

The *tormal* was a pleasant little animal, and Calhoun liked him very much. But *tormals* were crew-members of one-man Med ships because their metabolism was very similar to that of humans, but no *tormal* had ever been known to die of an infectious disease. They could play host to human infections, but only once and only lightly. It appeared that the furry little creatures had a hair-trigger sensitivity to bacterial toxins. The presence of infective material in their blood streams produced instant and violent reaction—and the production of antibodies in large quantity. Theorists said that *tormals* had dynamic immunity-systems instead of passive ones, like humans. Their body-chemistry seemed to look truculently for microscopic enemies to destroy, rather than to wait for something to develop before they fought it.

If he reacted normally, now, in a matter of hours his blood stream would be saturated with antibodies—or an antibody—lethal to the cultures Calhoun had injected. There was, however, one unfortunate fact. Murgatroyd weighed perhaps twenty pounds. There was most of a planetary population needing antibodies only he could produce.

He slept from breakfast-time to lunch. He breathed slightly faster than he should. His heart beat was troubled.

Calhoun swore a little when noon came. He looked at the equipment all laid out for biological micro-analysis—tiny test tubes holding half a drop, reagent flasks dispensing fractions of milliliters, tools and scales much tinier than doll-size. If he could determine the structure and formula of an antibody—or antibodies—that Murgatroyd's tiny body formed—why synthesis in quantity should be possible. Only the Med ship had not materials for so great an amount of product.

There was only one chance. Calhoun threw the spacephone switch. Instantly a voice came from the Speaker.

"... *ing Med Ship Esclipus Twenty! Phaedra fleet calling Med Ship Esclipus Twenty!*"

"Med ship answering," said Calhoun. "What is it?"

The voice went on: "*Calling Med Ship Twenty! Calling Med Ship Esclipus Twenty! Calling—*" It went on interminably. It was a very long way off, if it took so long for Calhoun's answer to be heard. But the call-formula broke off. "*Med ship! Our doctors want to know the trouble on Canis! Can we help? We've hospital ships equipped and ready!*"

"The question," said Calhoun steadily, "is whether I can make a formula-and-structure identification, and whether you can synthesize what I identify. How's your lab? How are you supplied with biological crudes?"

He waited. By the interval between his answer and a reply to it, the ship he'd communicated with was some five million miles or more away. But it

was still not as far as the next outward planet where the Phaedian fleet was based.

While he waited for his answer, Calhoun heard murmurings. They would come from the control building at the side of the grid. The loutish, suspicious gang there was listening. Calhoun had threatened to wreck the grid if they tried anything on the Med ship—but he could do nothing unless they tried to use a force-field. They listened in, muttering among themselves.

A long time later the voice from space came back. The fleet of the older generation of Phaedia was grounded, save for observation ships like the one speaking. The fleet had full biological equipment for any emergency. It could synthesize any desired compound up to—The degree of complexity and the classification was satisfactory.

"Day before yesterday," said Calhoun, "when you had me aground on Canis IV, your leader Walker said your children on this planet were destroying your grandchildren. He didn't say how. But the process is well under way—only the whole population will probably go with them. Most of the population, anyhow. I'm going to need those hospital ships and your best biological chemists—I hope! Get them started this way—fast! I'll try to make a deal for at least the hospital ships to be allowed to land. Over."

He did not flick off the spacephone. He listened. And a bitter, envenomed voice came from nearby:

"Sure! Sure! We'll let 'em land ships they say are hospital ships, loaded down with men and guns! We'll land 'em ourselves, we will!"

There was a click. The spacephone in the control building was turned off.

Calhoun turned back to the sleeping Murgatroyd. There was a movement about the grid-control building. Sleek, glistening ground-cars hurtled away—two of them. Calhoun turned then to the planetary communicator. It could break in on any wave-length used for radio communication under a planet's Heaviside roof. He had to get in touch with Walker or some other of the first-landed colonists. They were still embittered against their home world,

but they must be beginning to realize that Calhoun had told the truth about the youngest children. They'd find sickness if they looked for it.

But the planetary communicator picked up nothing. No radiation wavelength was in use. There was no organized news service. The young people on Canis III were too self-centered to care about news. There were no entertainment programs. Only show-offs would want to broadcast, and show-offs would not make the apparatus.

So Calhoun could not communicate save by spacephone, with a range of millions of miles, and the ship's exterior loud-speakers, with a range of hundreds of feet. If he left the Med ship, he wasn't likely to be able to fight his way back in. He couldn't find the younger Walker on foot, in any case, and he did not know anyone else to seek.

Besides, there was work to be done in the ship.

Before Murgatroyd waked, the two ground-cars had returned. At intervals, nearly a dozen other cars followed to the control building, hurtling across the grid's clear center with magnificent clouds of dust following them. They braked violently when they arrived. Youths piled out. Some of them yelled at the Med ship and made threatening gestures. They swarmed into the building.

Murgatroyd said tentatively, "*Chee?*"

He was awake. Calhoun could have embraced him.

"Now we see what we see!" he said grimly. "I hope you've done your stuff, Murgatroyd!"

Murgatroyd came obligingly to him, and Calhoun lifted him to the table he had ready. Again, what he did did not hurt. A tiny patch on Murgatroyd's side had been made permanently insensitive shortly after he was born. Calhoun extracted a quantity of what he hoped was a highly concentrated bacterial antagonist. He took thirty CCs in all. He clumped the red cells. He separated the serum. He diluted an infinitesimal bit of it and with a steady hand added it to a slide of the same cultures—living—on which Murgatroyd's dynamic immunity system had worked.

The cultures died immediately.

Calhoun had an antibody sample which could end the intolerable now-spreading disaster on the world of young people—if he could analyze it swiftly and accurately, and *if* the hospital ships from Phaedra could be landed, and *if* they could synthesize some highly complex antibody compounds, and *if* the inhabitants of Canis III would lay aside their hatred —

He heard a tapping sound on the Med ship's hull. He looked at a screen. Two youths stood in the doorway of the control building, leisurely shooting at the Med ship with sporting weapons.

Calhoun set to work. Sporting rifles were not apt to do much damage.

For an hour, while there was the occasional clanking of a missile against the ship's outer planking, he worked at the infinitely delicate job of separating serum from its antibody content. For another hour he tried to separate the antibody into fractions. Incredibly, it would not separate. It was one substance only.

There was a crackling sound and the whole ship shivered. The screen showed a cloud of smoke drifting away. The members of the grid-guard had detonated some explosive—intended for mining, most likely—against one of the landing-fins.

Calhoun swore. His call to the Phaedrian fleet was the cause. The grid-guard meant to allow no landing. He'd threatened to blow out their controls if they tried to use the grid on the Med ship, but they wanted it ready for use as a weapon against the space fleet. They couldn't use it against him. He couldn't damage it unless they tried. They wanted him away.

He went back to his work. From time to time, annoyedly, he looked up at the outside. Presently a young-warrior group moved toward the ship, carrying something very heavy. A larger charge of explosive, perhaps.

He waited until they were within yards of the ship. He stabbed the emergency-rocket button. A thin, pencillike rod of flame shot downward between the landing-fins. It was blue-white—the white of a sun's surface. For one instant it splashed out hungrily before it bored and melted a hole into the ground itself into which to flow. But in that instant it had ignited the covering of the burden the youths carried. They dropped it and fled. The

pencil flame bored deeper and deeper into the ground. Clouds of smoke and steam arose.

There was a lurid flash. The burden that the young warriors had abandoned, vanished in a flare that looked like a lightning bolt. The ship quivered from the detonation. A crater appeared where the explosive had been.



Calhoun cut off the emergency rocket, which had burned for ten seconds at one-quarter thrust.

Sunset came and night fell for the second time. He noticed, abruptly, that some of the ground-cars from about the control building went racing away. But they did not pass close to the Med ship in their departure. He labored on. He'd spent nearly thirty hours making cultures from the specimens swabbed from children's throats, and injecting Murgatroyd, and waiting for his reaction, and then separating a tiny quantity of antibody—which would

not total more than the dust from a butterfly's wing—from the serum he obtained.

Now he worked on, through the night. Far away—some tens or scores of millions of miles—the hospital ships of the Phaedrian fleet took off from the next outward planet. They would be coming at full speed toward Canis III. They would need the results of the work Calhoun was doing, if they were to prevent an appalling multiple plague which could wipe out all the sacrifice the building of the colony had entailed. But his work had to be exact.

It was tedious. It was exacting. It was exhaustingly time-consuming. He did have the help of previous experience, and the knowledge that the most probable molecular design would include this group of radicals and probably that, and side-chains like this might be looked for, and co-polymers might—But he was bleary-eyed and worn out before dawn came again. His eyes felt as if there were grains of sand beneath their lids. His brain felt dry—felt fibrous inside his skull, as if it were excelsior. But when the first red colors showed in the east, with the towers of the city against them, he had the blueprint of what should be the complex molecule formed in Murgatroyd's furry body.

He had just begun to realize, vaguely, that his work was done, when twin glaring lights came bouncing and plunging across the empty center of the grid. They were extraordinarily bright in the ruddy darkness. They stopped. A man jumped from the ground-car and ran toward the ship.

Calhoun wearily threw on the outer microphones and speakers.

"What's the matter now?" The man was the younger Walker.

"You're right!" called Walker's voice, strained to the breaking point. "There is sickness! Everywhere! There's an epidemic! It's just beginning! People felt tired and peevish and shut themselves away. Nobody realized! But they've got fevers! They're showing rashes! There's some delirium! The smallest children are worst—they were always quiet—but it's everywhere! We've never had real sickness before! What can we do?"

Calhoun said tiredly:

"I've got the design for an antibody. Murgatroyd made it. It's what he's for. The hospital ships from Phaedra are on the way now. They'll start turning it

out in quantity and their doctors will start giving everybody shots of it."

Young Walker cried out fiercely:

"But that would mean they'd land! They'd take over! I can't let them land! I haven't the power! Nobody has! Too many of us would rather die than let them land! They lied to us. It's bad enough to have them hovering outside. If they land, there'll be fighting everywhere and forever! We can't let them help us! We won't! We'll fight—we'll die first!"

Calhoun blinked, owlshly.

"That," he said exhaustedly, "is something you have to figure out for yourself. If you're determined to die, I can't stop you. Die first or die second—it's your choice. You make it. I'm going to sleep!"

He cut off the mike and speakers. He couldn't keep his eyes open.

VII.

"... As a strictly practical matter, a man who has to leave a task that he has finished, and wishes it to remain as he leaves it, usually finds it necessary to give the credit for his work to someone who will remain on the spot and will thereby be moved to protect and defend it so long as he lives...."

Manual, Interstellar Medical Service. Pp. 167-8.

Murgatroyd tugged at Calhoun and shrilled anxiously into his ear.

"*Chee-chee!*" he cried frantically. "*Chee-chee-chee!*"

Calhoun blinked open his eyes. There was a crashing sound and the Med ship swayed upon its landing-fins. It almost went over. It teetered horribly, and then slowly swung back past uprightness and tilted nearly as far in the opposite direction. There were crunching sounds as the soil partly gave way beneath one landing fin.



Then Calhoun waked thoroughly. In one movement he was up and launching himself across the cabin to the control-chair. There was another violent impact. He swept his hand across the row of studs which turned on all sources of information and communication. The screens came on, and the spacephone, and the outside mikes and loud-speakers, and even the planetary communication unit which would have reported had there been any use of the electromagnetic spectrum in the atmosphere of this planet.

Bedlam filled the cabin. From the spacephone speaker a stentorian voice shouted:

"This is our last word! Permit our landing or—"

A thunderous detonation was reported by the outside mikes. The Med ship fairly bounced. There was swirling white smoke outside the ship. It was mid-morning, now, and the giant lacy structure of the landing-grid was silhouetted against a deep-blue sky. There were cracklings from some electric storm perhaps a thousand miles away. There were shoutings, also brought in by the outside mikes.

Two groups of figures, fifty or a hundred yards from the Med ship, labored furiously over some objects on the ground. Smoke billowed out; then a heavy, blastlike *"Boom!"* Something came spinning through the air, end over end, with sputtering sparks trailing behind it. It fell close by the base of the upright Med ship.

Calhoun struck down the emergency rocket stud as it exploded. The roar of the rocket filled the interior of the ship. The spacephone speaker bellowed again:

"We've got a megaton-bomb missile headed down! This is our last word! Permit landing or we come in fighting!"

The object from the crude cannon went off violently. With the emergency rocket flaming to help, it lifted the Med ship, which jerked upward, settled back—and only two of its fins touched solidity. It began to topple because there was no support for the third.

Once toppled over, it would be helpless. It could be blasted with deliberately placed charges between its hull and the ground. A crater already existed where support for the third landing fin should have been.

Calhoun pushed the stud down full. The ship steadied and lifted. It went swinging across the level center of the landing-grid. Its slender, ultra-high-velocity flame knifed down through the sod, leaving a smoking, incandescent slash behind. The figures about the bomb-throwers scattered and fled. The Med ship straightened to an upright position and began to rise.

Calhoun swore. The grid was the planet's defense against landings from space, because it could fling out missiles of any size with perfect aim at any target within some hundred thousand miles—a good twelve planetary diameters. Its operators meant to defy the fleet from Phaedra and had to get

rid of the Med ship before they dared energize its coils. Now they were rid of it. Now they could throw bombs, or boulders, or anything else its force-fields could handle.

The spacephone roared again:

"On the ground there! Our missile is aimed straight for your grid! It carries a megaton fission bomb! Evacuate the area!"

Calhoun swore again. The gang, the guard, the young-warrior group at the grid would be far too self-confident to heed such a threat. If there were wiser heads on Canis III, they could not enforce their commands. A human community has to be complete or it is not workable. The civilization which had existed on Phaedra II was shattered by the coming doom of its sun. The fragments—on Phaedra, in the fleet, in each small occupied community on Canis III—were incomplete and incapable of thinking or acting in concert with any other. Every small group on this planet, certainly, gave only lip-service to the rest. The young world was inherently incapable of organizing itself, save on a miniature scale. And one such miniature group had the grid and would fight with it regardless of the wishes of any other—because that group happened to be composed of instinct-driven members of the young-warrior group.

But he was still within the half-mile-high fence of the grid's steel structure. He strapped himself in his seat. The ship rose and rose. It came level with the top of the colony's one defense against space. The peculiar, corrugated copper lip of that structure, formed into the force-field guide which made it usable, swung toward him. He raised the rocket-thrust and shot skyward.

A deafening bellow came from the speakers:

"Yeah! Go on out and join the old folk! We'll get you!"

Obviously, the voice was from the ground below him. The ship flashed upward. Calhoun rasped into the spacephone mike, himself:

"Med ship calling fleet! Call back that missile! I've got the antibody structure! This is no time for fighting! Call your missile back!"

Derisive laughter—again from the ground. Then the heavy, growling voice of an older man.

"Keep out of the way, Med ship! These young fools are destroying themselves. Now they're destroying our grandchildren! If we hadn't been soft-hearted before—if we'd fought them from the beginning—the little ones wouldn't be dying now! Keep out of the way! If you can help us, it'll be after we've won the war!"

The sky turned purple, at the height Calhoun had reached. It went black. The sun Canis flamed and flared against a background of ebony space, sprinkled with a thousand million colored stars. The Med ship continued to rise.

Calhoun felt singularly and helplessly alone. Below him the sunlit surface of a world spread out, its edge already curving, cloud-masses in its atmosphere veiling the details of mountains and green-clad plains. There was the blue of ocean, creeping in. The city of the landing-grid was tiny, now. The brown of ploughed fields was no longer divided into rectangular shapes. It was a mere brownish haze between the colorings of as-yet-untouched virgin areas. The colonists of Canis III had so far made only a part of the new world their own. Many times more remained to be turned to human use.

The rear screen showed something coming upward. Masses of stuff, without shape but with terrific velocity. It was inchoate, indefinite stuff. It was plain dirt from the center of the landing-grid's floor, flung upward with the horrible power available for the landing and launching of ships. And, focused upon it, the force-fields of the grid could control it absolutely for a hundred thousand miles.

Calhoun swerved, ever so slightly. His own velocity had reached miles per second, but the formless mass following him was traveling at tens. It would not matter what such a hurtling missile was. At such a velocity it would not strike like a mass, but like a meteor-shower, flaring into incandescence when it touched and vaporizing the Med ship with itself in the flame of impact.

But the grid would have to let go before it hit. There was monstrous stored power in the ship's Duhanne cells. If so much raw energy were released into

anything on which a force-field was focused, it would destroy the source of the field. The grid could control its battering-ram until the very last fraction of a second, but then it must release—and its operator knew it.

Calhoun swung his ship frantically.

The mass of speeding planet-matter raced past no more than hundreds of yards away. It was released. It would go on through empty space for months or years—perhaps forever.

Calhoun swung back to his upward course. Now he sent raging commands before him:

"Pull back that missile! You can't land a bomb on Canis! There are people there! You can't drop a bomb on Canis!"

There was no answer. He raged again:

"Med ship calling Phaedra fleet! There's disease on Canis! Your children and grandchildren are stricken! You can't fight your way to help them! You can't blast your way to sickbeds! You've got to negotiate! You've got to compromise! You've got to make a bargain or you and they together—"

A snarling voice from the ground said spitefully:

"Never mind, little Med man! Let 'em try to land! Let 'em try to take over and boss us! We listened to them long enough! Let 'em try to land and see what happens! We've got their fleet spotted! We'll take care of them!"

Then the growling tones Calhoun had come to associate with Phaedra:

"You keep out of the way, Med Ship. If our young children are sick, we're going to them! We're just beyond the area in which no drive will work. When the grid has been blasted our landing ship will go down and we'll come in! Our missile is only half an hour from target now! We'll begin our landing in three hours or less! Out of the way!"

Calhoun said very bitter and extremely impolite words. But he faced an absolute emotional stalemate between enemies of whom both were in the wrong. The frantic anger of the adults of Phaedra, barred from the world to

which they'd sent their children first so they could stay where doom awaited, was matched by the embittered revolt of the young people who had been worked past endurance and burdened past anyone's power to tolerate. There could be no compromise. It was not possible for either side to confess even partial defeat by the other. The quarrel had to be fought to a finish as between the opposing sides, and then hatred would remain no matter which side won. Such hatred could not be reasoned with.

It could only be replaced by a greater hatred.

Calhoun ground his teeth. The Med ship hurtled out from the sunlit Canis III. Somewhere—not many thousands of miles away—the fleet of Phaedra clustered. Its crews were raging, but they were sick with anxiety about the enemies they prepared to fight. Aground there was hatred among the older of the colonists—the young-warrior group in particular, because that is the group in which hate is appropriate—and there was no less a sickish disturbance because even in being right they were wrong. Every decent impulse that had been played upon to make them exhaust themselves, before their revolt, now protested the consequences of their revolt. Yet they believed that in revolting they were justified.

Murgatroyd did not like the continued roar of emergency-rockets. He climbed up on Calhoun's lap and protested.

"*Chee!*" he said urgently. "*Chee-chee!*"

Calhoun grunted.

"Murgatroyd," he said, "it is a Med service rule that a Med ship man is expendable in case of need. I'm very much afraid that we've got to be expended. Hang on, now! We try some action!"

He turned the Med ship end for end and fed full power to the rockets. The ship would decelerate even faster than it had gathered speed. He set the nearest-object indicator to high gain. It showed the now-retreating mass of stone and soil from Canis. Calhoun then set up a scanner to examine a particular part of the sky.

"Since fathers can be insulted," he observed, "they've made a missile to fight its way down through anything that's thrown at it. It'll be remote-controlled for the purpose. It's very doubtful that there's a spaceship on the planet to fight it back. There's been no reference to one, anyhow. So what the missile will have to fight off will be stuff from the landing-grid only. Which is good. Moreover, fathers being what they are, regardless, that missile won't be a high-speed one. They'll want to be able to call it back at the last minute. They'll hope to."

"*Chee!*" said Murgatroyd, insisting that he didn't like the rocket-roar.

"So we will make ourselves as unpopular as possible with the fathers," observed Calhoun, "and if we live through it we will make ourselves even more cordially hated by the sons. And then they will be able to tolerate each other a little, because they both hate us so much. And so the public-health situation on Canis III may be resolved. Ah!"

The nearest-object indicator showed something moving toward the Med ship. The scanner repeated the information in greater detail. There was a small object headed toward the planet from empty space. Its velocity and course—

Calhoun put on double acceleration to intercept it, while he pointed the ship quartering so he'd continue to lose outward speed.

Ten minutes later the spacephone growled:

"Med ship! What do you think you're doing?"

"Getting in trouble," said Calhoun briefly.

Silence. The screens showed a tiny pin point of moving light, far away toward emptiness. Calhoun computed his course. He changed it.

"Med ship!" rasped the spacephone. *"Keep out of the way of our missile! It's a megaton bomb!"*

Calhoun said irrelevantly:

"Those who in quarrels interpose, must often wipe a bloody nose." He added. "I know what it is."

"Let it alone!" rasped the voice. *"The grid on the ground has spotted it. They're sending up rocks to fight it."*

"They're rotten marksmen," said Calhoun. "They missed me!"

He aimed his ship. He knew the capacities of his ship as only a man who'd handled one for a long time could. He knew quite exactly what it could do.

The rocket from remoteness—the megaton-bomb guided missile—came smoking furiously from the stars. Calhoun seemed to throw his ship into a collision course. The rocket swerved to avoid him, though guided from many thousands of miles away. There was a trivial time-lag, too, between the time its scanners picked up a picture and transmitted it, and the transmission reached the Phaedrian fleet and the controlling impulses reached the missile in response. Calhoun counted on that. He had to. But he wasn't trying for a collision. He was forcing evasive action. He secured it. The rocket slanted itself to dart aside, and Calhoun threw the Med ship into a flip-flop and—it was a hair-raising thing—slashed the rocket lengthwise with his rocket flame. That flame was less than half an inch thick, but it was of the temperature of the surface of a star, and in emptiness it was some hundreds of yards long. It sliced the rocket neatly. It flamed hideously, and even so far, Calhoun felt a cushioned impact from the flame. But that was the missile's rocket fuel. An atom bomb is the one known kind of bomb which will not be exploded by being sliced in half.

The fragments of the guided missile went on toward the planet, but they were harmless.

"*All right!*" said the spacephone icily—but Calhoun thought there was relief in the voice. "*You've only delayed our landing and lost a good many lives to disease!*"

Calhoun swallowed something he suspected was his heart, come up into his throat.

"Now," he said, "we'll see if that's true!"

His ship had lost its spaceward velocity before it met the missile. Now it was gaining velocity toward the planet. He cut off the rocket to observe. He swung the hull about and gave a couple of short rocket blasts.

"I'd better get economical," he told Murgatroyd. "Rocket fuel is hard to come by, this far out in space. If I don't watch out, we'll be caught in orbit, here, with no way to get down! I don't think the local inhabitants would be inclined to help us."

His lateral dash at the missile had given him something close to orbital speed relative to the planet's surface, though. The Med ship went floating, with seemingly infinite leisure, around the vast bulge of the embattled world. In less than half an hour it was deep in the blackness of Canis' nighttime shadow. In three-quarters of an hour it came out again at the sunrise edge, barely four hundred miles high.

"Not quite speed enough for a true orbit," he told Murgatroyd critically. "I'd give a lot for a good map!"

He watched alertly. He could gain more height if he needed to, but he was worried about rocket fuel. It is intended for dire emergencies only. It weighed too much to be carried in quantity.

He spotted the city of Canopolis on the horizon. He became furiously busy. He inverted the little ship and dived down into atmosphere. He killed speed with rocket flames and air friction together, falling recklessly the while. He was barely two miles high when he swept past a ridge of mountains and the city lay ahead and below. He could have crashed just short of it. But he spent more fuel to stay aloft. He used the rockets twice. Delicately.

At a ground speed of perhaps as little as two hundred miles an hour, supported at the end by a jetting, hair-thin rocket flame that was like a rod of electric arc-fire, he swept across the top of the landing-grid. The swordlike flame washed briefly over the nearer edge. Very briefly. The flame cut a slash down through steel girders and heavy copper cables together. The rockets roared furiously. That one disabling cut at the grid had been on a downward, darting drift. Now the ship shouted, and swooped up, and on—and it swept above the far side of the grid only yards from the wide strip of copper which guided its force-fields out into space. Here it cut cables, girders, and force-field guide together for better than two hundred feet from the top. The grid was useless until painstaking labor had made the damage good.

Calhoun used nearly the last of his fuel for height while he said crisply on the spacephone:

"Calling fleet! Calling fleet! Med ship calling fleet! I've disabled the landing-grid on Canopolis! You can come in now and take care of the sick. There are no weapons aground to speak of and if you don't get trigger-happy there should be no fighting. I'll be landed off somewhere in the hills to the north of the town. If the local inhabitants don't pack explosives out and crack the ship to get at me, I'll have the facts on the antibody ready for you. In fact, as soon as I get down I'll give them to you by spacephone, just in case."

It was a near thing, though. His rocket fuel was exhausted when he hit the ground. The flame sputtered and stopped when the ship was three feet from touching. It fell over, splintering trees. It was distinctly a rough landing.

Murgatroyd was very indignant about it. He scolded shrilly while Calhoun unstrapped himself from the chair and when he looked out to see where they were.

It was a week later when the Med ship—brought to the grid for repair and refueling—was ready for space again. The original landing-grid still stood, of course. But it was straddled and overwhelmed, huge as it was, by the utterly gigantic flying grid from Phaedra. There were not many ships aground, though. As Calhoun moved toward the control building, now connected by cable to the control quarters in the flying grid, one of the few ships remaining seemed to fall toward the sky. A second ship followed only seconds later.

He went into the control building. Walker the elder, from Phaedra, nodded remotely as he entered. The younger Walker scowled at him. He had been in consultation with his father, and the atmosphere was one of great reserve.

"Hm-m-m," said the elder Walker, gruffly. "What's the report?"

"Fairly good," said Calhoun. "There was one lot of antibody that seems to have been a trifle under strength. But the general situation seems satisfactory. There'll be a few more cases of one thing and another, of

course—cases that are incubating now. But they'll do all right on the antibody shots. They have so far, at any rate." He said to the younger Walker, "You did a very good job rounding up the thirteen-to-fifteen-year-olds to escort the fleet doctors and handle the patients for them. They took themselves very seriously. They were ideal for the job. Your young-warrior group—"

"A lot of them," said young Walker dourly, "have taken to the woods. They swear they'll never give in!"

"How about the girls?"

Young Walker shrugged.

"They're fluttering about and beginning to talk about clothes. When older women arrive there'll be dress-making—"

"And the lads in the woods," said Calhoun, "will come out to fascinate, and be fascinated instead. Do you think there'll be really much trouble?"

"No-o-o," said young Walker sourly. "Some of ... our younger crowd seem relieved to be rid of responsibility."

"But," interposed the older Walker, gruffly, "he wants it. He thrives on it. He'll get it!" He hrrumphed. "The same with the others who showed what they could do here. We oldsters need them. We don't plan any ... ah ... reprisals."

Calhoun raised his eyebrows.

"Should I be surprised?"

The older Walker snorted.

"You didn't expect us to fall into each other's arms after what's happened, did you? No! But we are going to try to ignore our ... differences as much as we can. We won't forget them, though."

"I suspect," said Calhoun, "that they'll be harder to remember than you think. You had a culture that split apart. Its pieces were incomplete—and a society has to be complete to survive. It isn't a human invention. It's something we have an instinct for—as birds have an instinct to build nests. When we build a culture according to our instincts, we get along. When

that's impossible—there's trouble." Then he said, "I'm not trying to lecture you."

"Oh," said the elder Walker. "You aren't?"

Calhoun grinned.

"I thought I'd be the most unpopular man on this planet," he said cheerfully. "And I am. I interfered in everybody's business and nobody carried out his plans the way he wanted to. But at least nobody feels like he won. You'll be pleased when I lift the quarantine and take off, won't you?"

The older Walker said scornfully:

"We're paying no attention to your quarantine! Our fleet's loading up our wives on Phaedra, to ferry them here as fast as overdrive will do it! D'you think we'd pay any attention to your quarantine?"

Calhoun grinned again. The younger Walker said painfully:

"I suppose you think we should—" He stopped, and said very carefully: "What you did was for our good, all right, but it hurts us more than it does you. In twenty years, maybe, we'll be able to laugh at ourselves. Then we'll feel grateful. Now we know what we owe you, but we don't like it."

"And that," said Calhoun, "means that everything is back to normal. That's the traditional attitude toward all medical men—owe them a lot and hate to pay. I'll sign the quarantine release and take off as soon as you give me some rocket fuel, just in case of emergency."

"Right away!" said the two Walkers, in unison.

Calhoun snapped his fingers. Murgatroyd swaggered to his side. Calhoun took the little *tormal*'s black paw in his hand.

"Come along, Murgatroyd," he said cheerfully. "You're the only person I really treated badly, and you don't mind. I suppose the moral of all this is that a *tormal* is a man's best friend."

*** END OF THE PROJECT GUTENBERG EBOOK THE
GRANDFATHERS' WAR ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.